

Body-to-brain insulin and Notch signaling regulates memory through neuronal CREB activity

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While memory regulation is predominantly understood as autonomous to neurons, factors outside the brain can also affect neuronal function. In *Caenorhabditis elegans*, the insulin/IGF-1-like signaling (IIS) pathway regulates longevity, metabolism and memory: long-lived *daf-2* insulin/IGF-1 receptor mutants more than double memory duration after a single training session, and it was assumed that memory regulation was strictly neuronal. However, here we show that degradation of DAF-2 in the hypodermis also greatly extends memory, via expression of the diffusible Notch ligand, OSM-11, which in turn activates Notch signaling in neurons. Single-nucleus RNA sequencing of neurons revealed increased expression of CREB and other memory genes. Furthermore, in aged animals, activation of the hypodermal IIS–Notch pathway as well as OSM-11 overexpression rescue both memory and learning via CREB activity. Thus, insulin signaling in the liver-like hypodermis non-autonomously regulates neuronal function, providing a systemic connection between metabolism and memory through IIS–Notch–CREB signaling from the body to the brain.

Memory loss is among the most debilitating symptoms of aging. As human lifespan continues to increase¹, a challenge must be addressed: how can we slow cognitive aging? Age-related cognitive decline starts in middle age², with learning and memory decreasing more rapidly than other behavioral declines, even in the absence of disease^{3,4}. While mechanisms underlying memory regulation and cognitive aging have been studied for decades, therapies for age-related cognitive decline are still lacking. Neuron-based mediators of memory have been the primary focus, as the brain is the functional location of cognition and memory. However, the role of peripheral tissues on brain function is becoming appreciated. For example, mouse parabiosis experiments revealed blood factors that influence cognitive function^{5,6}. Whether any tissues outside neurons play a similar role in the regulation of *Caenorhabditis elegans* memory, and therefore might act as possible

targets of midlife or late-life interventions to slow cognitive decline, is underexplored.

The molecular machinery required to form and maintain short-term and long-term memories is conserved from *C. elegans* to mammals^{7,8}. For example, pathways controlling presynaptic vesicle transport and release are vital components regulating memory maintenance in both worms and humans^{9–12}. Like mammals, *C. elegans* can learn and form both short-term and long-term associative memories^{3,8}; memory is measured using a Pavlovian-style appetitive associative learning and memory assay³. Young adult wild-type worms retain short-term memory for 1–2 h³; short-term memory decreases with age and is lost in midlife³. Long-term memory induced by multiple training sessions persists for 16–24 h and requires gene transcription by the cAMP response element-binding protein (CREB)³, which is activated

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in a single pair of interneurons (AIM) upon spaced training⁷. Long-term associative memory (LTAM) ability is lost even earlier in adulthood than short-term memory, correlating with declining CREB levels³. CREB is a conserved regulator of long-term memory across organisms, including *Aplysia*, *C. elegans*, *Drosophila*, and mice⁷. Overexpression of CREB specifically in neurons³ or gain-of-function in Gαq/GNAQ signaling upstream of CREB¹³ each enhances worms' long-term memory, including in aged worms, and activation of Gαq/GNAQ in aged mice restores memory¹². CREB expression decreases with age both in *C. elegans*³ and in mammals¹⁴, a trend paralleling age-related memory decline^{3,14}. This implies that sustaining or restoring CREB levels/activity might present a strategy to prevent or slow cognitive aging.

The IIS longevity pathway is critical for regulation of aging in worms through mammals⁹. This pathway both autonomously and non-autonomously regulates longevity¹⁵, metabolism^{16,17} and reproduction^{18–22}. Additionally, DAF-2 plays tissue-specific roles in longevity and dauer formation regulation²³. We previously found that reducing IIS via mutation of the insulin/IGF-like receptor, *daf-2*, significantly extends associative memory and slows cognitive aging³. We also identified genes responsible for IIS regulation of this memory improvement through transcriptomic analysis of neurons isolated from *daf-2* mutants¹⁰. However, a non-autonomous role for IIS in regulation of cognitive function has not previously been explored.

While investigating tissue-specific functions of IIS activity using the auxin-inducible degron (AID) protein degradation system to selectively degrade DAF-2 (ref. 23), we made the surprising discovery that loss of DAF-2 in the hypodermis, the metabolic epidermal tissue of the worm²⁰, significantly extends memory. That is, in addition to the expected memory improvement by DAF-2 degradation in neurons, hypodermal DAF-2 degradation significantly improves memory. We then found that the hypodermis signals to neurons via the OSM-11 Notch ligand and Notch signaling components in neurons. Using single-nucleus RNA sequencing (RNA-seq) of neurons, we identified transcriptional changes in individual neurons downstream of hypodermal IIS signaling, revealing that hypodermal DAF-2 degradation broadly induces expression of CREB, and hypodermal DAF-2's extended memory is dependent on CREB. Unlike hypodermal DAF-2 depletion memory extension, the memory enhancement caused by DAF-2 degradation in neurons is independent of CREB. We further established that memory regulation by the hypodermis is mediated by Notch ligand signaling from the hypodermis to neurons and is dependent on CREB. Finally, reduction of hypodermal IIS or increased Notch signaling via OSM-11 expression in aged worms can each rescue memory decline. Our results suggest that the memory extension induced by the reduction of IIS in *daf-2* mutants is the result of the combination of CREB-dependent and CREB-independent gene expression changes regulated by hypodermal and neuronal IIS, respectively. These data

establish a role for non-autonomous regulation of CREB-dependent memory and cognitive maintenance by hypodermal IIS signaling and downstream Notch signaling. Thus, memory can be extended by insulin signaling in both the brain and the body, and DAF-2 activation rescues memory function in aged animals.

Results

Reduced hypodermal or neuron IIS extends associative memory

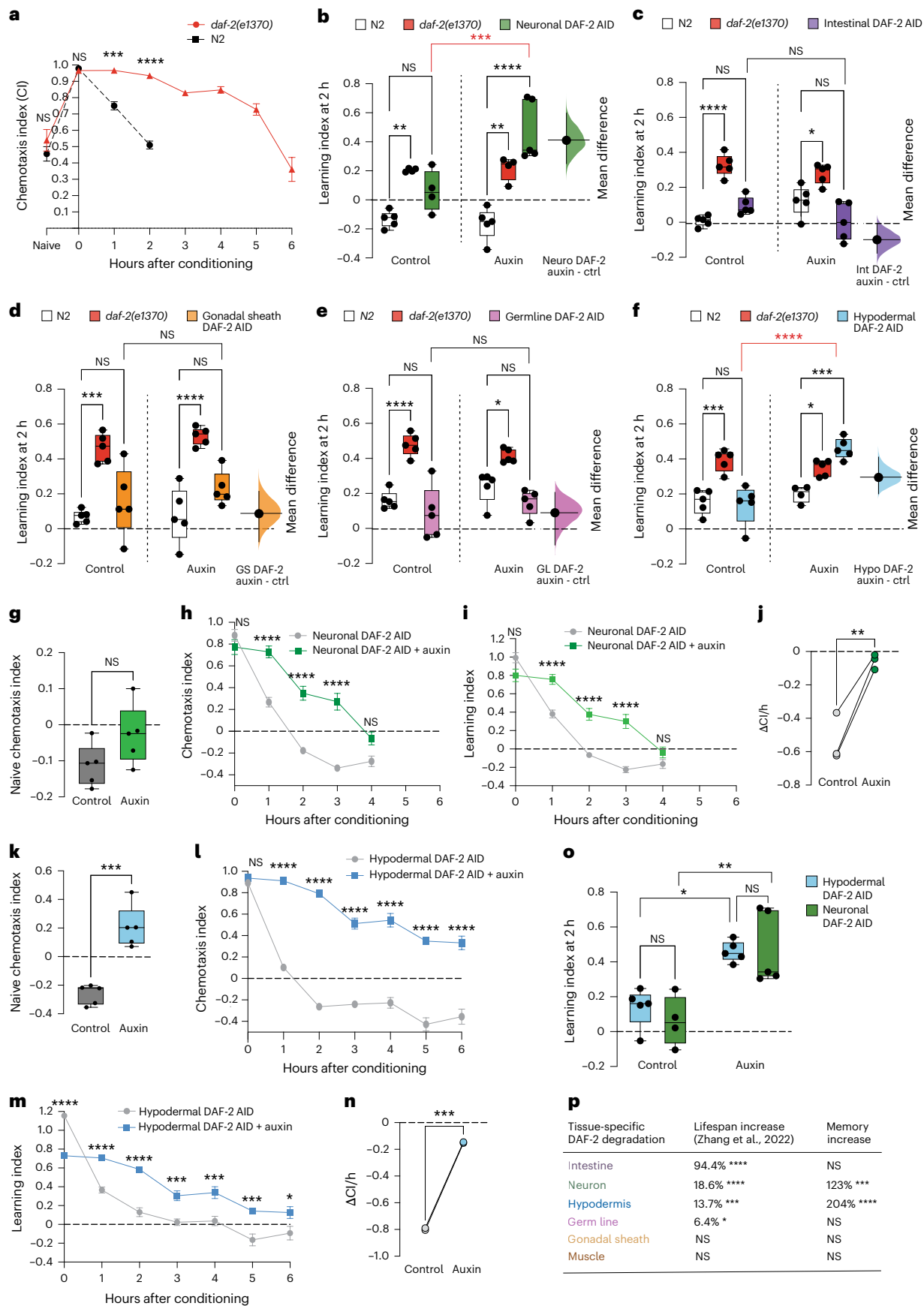
We previously found that reduction of IIS improves *C. elegans* associative memory in both young and aged adults and extends memory beyond 5 h after a single training session³. To test memory, we use a Pavlovian positive olfactory association assay³: short starvation followed by conditioning (training) of the animals with food + butanone (a neutral odorant) increases the worms' preference for butanone, indicating learning of this positive food–butanone association; holding the trained animals on food without butanone, then testing their chemotaxis to butanone at later times, measures associative memory duration (Extended Data Fig. 1a,b and Extended Data Table 1). We measure chemotaxis before training ('naive') to ensure that the chemotaxis index (CI) is low (−0.2 for wild type), indicating low stress. During the naive chemotaxis assay, we also assess whether there are any motility issues that would affect interpretation of the remainder of the assay; if the worms cannot move away from the origin, we do not perform learning and memory assays on those animals, to avoid confusing high naive preference with learning or memory. Immediately after conditioning (odor + food training), we assess the trained CI; learning ($t = 0$ after training) is the difference between naive and trained CI (Extended Data Fig. 1a,b). Worms' preference for butanone is tested at selected time points after a single training session (short-term associative memory (STAM); Extended Data Fig. 1a,b). Thus, we can distinctly assess each function (motility, naive butanone attraction, associative learning and associative memory). Consistent with our previous results, the insulin-like receptor mutant *daf-2* improves memory maintenance, retaining memory for more than 5 h after conditioning, while wild-type worms have forgotten the association between food and butanone by 2 h³ (Fig. 1a and Extended Data Fig. 1d,e).

Zhang et al. used the AID degradation system to selectively degrade DAF-2 in individual tissues to determine *daf-2*'s tissue-specific roles in the regulation of lifespan, dauer, and RNA and protein metabolism²³. This system allows DAF-2 degradation with temporal control and only in specific tissues in adult animals²³, avoiding any possible developmental effects due to systemically reduced IIS. The authors noted that DAF-2 is difficult to visualize due to its localization on the plasma membrane; therefore, they used GFP fusions to verify that DAF-2 was degraded in the specific tissues without affecting DAF-2 in other tissues in these TIR-1 lines²³.

Fig. 1 | Reduced insulin signaling in the hypodermis and neurons extends associative memory.

a, Young adult (day 1) *daf-2(e1370)* mutants have extended memory up to 6 h after one training session, while N2 wild-type worms' STAM lasts for <2 h. **b**, Auxin-induced DAF-2 degradation in neurons increases associative memory in day 2 adults. **c–e**, Auxin-induced DAF-2 degradation in other tissues (intestine (**c**); gonadal sheath (**d**); germ line (**e**)) has no effect on memory. **f**, Auxin-induced DAF-2 degradation in the hypodermis improves memory in young adults. **b–f**, Auxin treatment started 20 h before memory training. In the estimation plots, mean difference (bootstrap sampling distribution) is depicted as a dot. 95% confidence interval indicated by the vertical bar. **g–i**, Naive CI (**g**), CI (**h**) and learning index (**i**) of neuronal DAF-2-AID worms. **h,i**, Neuronal DAF-2 degradation extends memory up to 3 h. **j**, Memory was evaluated by calculating the rate of change in CI between the learning and 1-h time points across replicates. **k**, Naive chemotaxis and chemotaxis and learning index plots of hypodermal DAF-2-AID worms. **l,m**, Hypodermal DAF-2 degradation extends memory up to 6 h after a single training session, chemotaxis index (**l**) and learning index (**m**) of hypodermal DAF-2-AID worms. **n**, Slope

analysis calculated as in **j**. **o**, Comparison of memory improvement induced by hypodermal or neuronal DAF-2 degradation. **p**, Lifespan and memory increases upon tissue-specific DAF-2 degradation. Lifespan changes and *P* values reported from ref. 23. Memory increase is reported as the average increase in median memory relative to controls across all biological replicates. Memory *P* values from 2-h learning index. **b,f**, Red indicates comparison of note for genetic interpretation. **b–f**, Representative experiments from two to three biological replicates. Each dot indicates individual chemotaxis assay plate containing ~80 worms. Two-way repeated-measures analysis of variance (ANOVA) and Tukey post hoc tests were performed. In the box plots, the center line denotes the median, box range indicates the 25th–75th percentile, and whiskers denote minimum–maximum values. **a,h,i,l,m**, Representative experiment from three biological replicates. Mean ± s.e.m. Two-way repeated-measures (ANOVA) and Bonferroni post hoc tests were performed. $n \geq 4$ per time point. **g,k,j,n**, Two-sided unpaired *t*-test. NS, not significant, * $P \leq 0.05$, ** $P \leq 0.01$, *** $P \leq 0.001$, **** $P \leq 0.0001$.



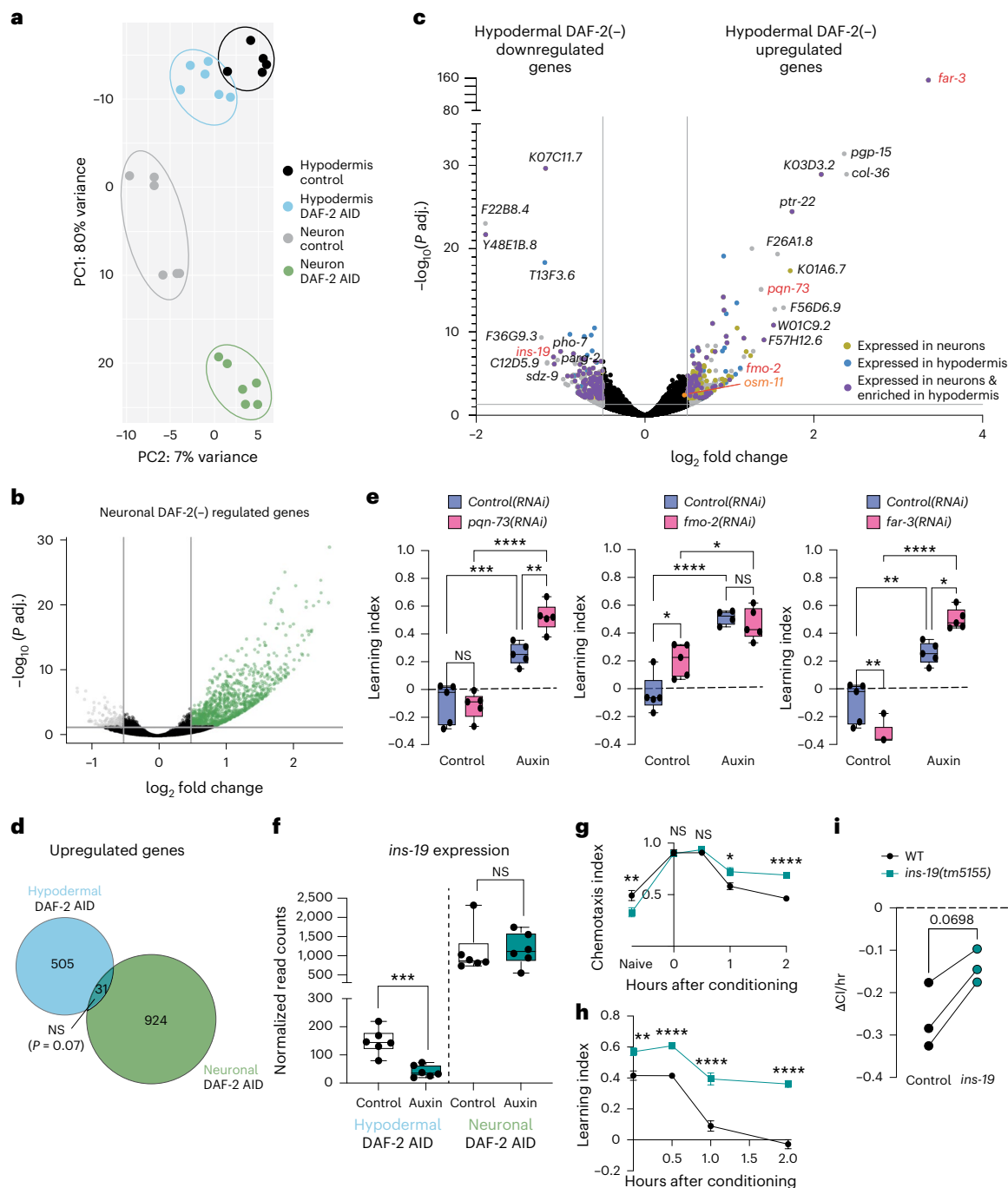


Fig. 2 | Hypodermal DAF-2(-) transcriptional targets are distinct from neuronal DAF-2(-) targets. a, PCA of mRNA-seq from whole worms with hypodermal DAF-2 degradation and neuronal DAF-2 degradation. Day 2 adult worms; six biological replicates per condition. **b**, Volcano plot of neuronal DAF-2(-)-regulated genes. Green dots indicate upregulated genes ($\log_2$ fold change ≥ 0.5). Gray dots indicate downregulated genes ($\log_2$ fold change ≤ -0.5). False discovery rate (FDR) $\leq 1\%$. **c**, Volcano plot of hypodermal DAF-2(-) upregulated and downregulated genes (gray dots, $\log_2$ fold change ≥ 0.5 or ≤ -0.5 , FDR $\leq 1\%$). Gold dots indicate genes expressed in neurons. Blue dots indicate genes expressed in the hypodermis. Purple dots indicate genes expressed in neurons and enriched in the hypodermis^{20,24}. **d**, Comparison of upregulated genes from neuronal and hypodermal DAF-2(-) animals. All upregulated genes, FDR $< 1\%$. Hypergeometric test. **e**, RNAi knockdown (from young adult (L4)) of the hypodermal DAF-2-regulated transcriptional targets *pqn-73*,

fmo-2, or *far-3* does not block memory enhancement induced by hypodermal DAF-2 degradation in young adults (tested on day 3). One biological replicate. **f**, *ins-19* expression is downregulated upon hypodermal DAF-2 degradation, but not upon neuronal DAF-2 degradation. Statistics from DESeq2 (adjusted P values). **g**, **h**, **i** (**g**) and learning index (**h**) plots of *ins-19* mutant. Mean $\pm$ s.e.m. Two-way repeated-measures ANOVA, Bonferroni post hoc tests. $n \geq 4$ per time point. n represents the number of chemotaxis plates at each time point. Each plate contains ~80 worms for all memory experiments. **i**, Memory was evaluated by calculating the rate of change in CI between the learning and 1-h time points across replicates. Two-sided unpaired t -test. **e**, **f**, Two-way repeated-measures ANOVA. In the box plots, the center line denotes the median, box range indicates the 25th–75th percentile, and whiskers denote minimum–maximum values. Representative experiment for two to three biological replicates unless specified otherwise. * $P \leq 0.05$, ** $P \leq 0.01$, *** $P \leq 0.001$, **** $P \leq 0.0001$.

We treated these DAF-2 AID worm strains with auxin on day 1 of adulthood, then carried out our single-training associative memory assay on day 2 (young adults; Extended Data Fig. 1c) after DAF-2 was degraded in neurons, intestine, gonadal sheath, germ line, muscle, or hypodermis (Fig. 1b–f and Extended Data Fig. 1f–k). As expected, neuronal DAF-2 degradation significantly extended memory duration (Fig. 1b and Extended Data Figs. 1f and 2a), confirming that the auxin treatment successfully degrades DAF-2, consistent with the role of neuron-autonomous IIS in memory and behavior¹⁰. Degradation of DAF-2 in the intestine, gonadal sheath, or germ line had no effect on memory (Fig. 1c–e and Extended Data Figs. 1g–i, k and 2b–d). The muscle DAF-2 AID strain shows significant defects in learning before AID treatment (Extended Data Fig. 1k), complicating interpretation. Surprisingly, however, we found that DAF-2 degradation specifically in the hypodermis dramatically increases memory duration (Fig. 1f and Extended Data Figs. 1j and 2e). The magnitude of memory improvement induced by hypodermal DAF-2 degradation is similar to that of neuronal DAF-2 degradation (Fig. 1b and Supplementary Table 1).

Next, we examined the duration of memory induced by neuronal and hypodermal degradation of DAF-2. Surprisingly, while auxin-induced neuronal DAF-2 degradation extended memory beyond 3 h after training (Fig. 1g–j), hypodermal DAF-2 degradation extended memory beyond 6 h after training (Fig. 1k–o), similar to the memory duration of *daf-2* mutants (Fig. 1a and Extended Data Fig. 1d).

To determine whether memory improvement is solely the consequence of increased lifespan, we compared the extension of lifespan by DAF-2 degradation in a specific tissue²³ and memory improvement (Fig. 1p). We found no correlation between extension of lifespan by DAF-2 degradation in a specific tissue and the effect on memory; for example, hypodermal DAF-2 degradation has the highest mean memory increase, but causes only a modest lifespan increase²³, while DAF-2 degradation in the intestine nearly doubles lifespan and, like *daf-2* mutants, increases dauer formation²³, as we verified (100% dauer; Extended Data Fig. 2h) but has no effect on memory (Fig. 1c and Extended Data Fig. 1g and 2a). These results eliminate the hypothesis that simply extending lifespan increases memory, suggesting that lifespan and memory are regulated by DAF-2 through distinct mechanisms.

Identification of hypodermal IIS transcriptional targets

To uncover the molecular mechanisms of non-autonomous memory regulation from hypodermal IIS, we identified downstream transcriptional targets regulated by hypodermal IIS using whole-worm RNA-seq. Comparison of the transcriptional profiles of day 2 non-auxin control and hypodermal DAF-2-degraded (+ auxin) worms revealed distinct transcriptional profiles (principal component analysis (PCA); Fig. 2a). Similarly, neuronal DAF-2-regulated genes (Fig. 2b) are distinct from hypodermal DAF-2-regulated genes (Fig. 2c,d). Although we manipulated DAF-2 in the hypodermis, most genes upregulated by hypodermal

IIS reduction are enriched in the nervous system and neurons, followed by the intestine and hypodermis (Extended Data Fig. 3a), consistent with our behavioral results (Figs 1 and 2c and Supplementary Table 2).

To assess the roles of genes downstream of hypodermal IIS, we knocked down the top upregulated genes and tested memory. *far-3*, the gene with the highest fold change (Fig. 2c), encodes a fatty acid binding protein that is expressed in the hypodermis, intestine and neurons, and is regulated by Notch signaling key components^{20,24}. *pqn-73* encodes a prion-like protein; we previously found that *pqn-73* is expressed in arcade cells and neurons, and is required for LTAM⁷. *fmo-2* encodes a flavin-containing monooxygenase expressed in the hypodermis and neurons, and activation of *fmo-2* in *C. elegans* extends lifespan²⁵. While *far-3* has DAF-16 binding elements (DBEs) and one DAF-16 associative element (DAE) in its upstream promoter, *pqn-73* and *fmo-2* have no DBEs or DAEs in their promoters and are thus less likely to be direct targets of the insulin signaling pathway in the hypodermis (Supplementary Table 3). However, knockdown of *far-3*, *pqn-73*, or *fmo-2* did not block memory enhancement induced by hypodermal DAF-2 degradation; in fact, *pqn-73* and *far-3* knockdown each slightly increased memory (Fig. 2e and Extended Data Fig. 3b–d), suggesting these genes were not responsible for the improved short-term memory induced by hypodermal IIS reduction. We next focused on the hypodermal DAF-2(–) downregulated insulin-like peptide, *ins-19*, which is downregulated only upon hypodermal DAF-2(–) conditions and not in neuronal DAF-2(–) animals (Fig. 2f). Because *ins-19* mutants have significantly higher learning and memory, consistent with previous results²⁶ (Fig. 2g–i and Extended Data Fig. 3e), hypodermis IIS-induced memory extension may be mediated in part by *ins-19* reduction.

Hypodermal-dependent memory requires the OSM-11 Notch ligand

We next sought to determine how the hypodermis—a structural and metabolic tissue—regulates memory, which is largely thought to be a neuron-autonomous process. We hypothesized that hypodermal IIS targets might include a secreted, diffusible protein that signals to neurons. We cross-referenced hypodermis-specific gene expression in adult *C. elegans*²⁰ with genes that are upregulated in *daf-2* mutants compared to *daf-16*;*daf-2* worms (class I genes)^{16,17}; 15 class I candidate genes are expressed in the hypodermis and are predicted to encode secreted proteins (Fig. 3a). Among those 15 genes, *osm-11* is expressed in the hypodermis^{27,28} and has the highest fold-induction following adult hypodermal DAF-2 degradation (Supplementary Table 2 and Fig. 3b; $P < 0.0034$; $\log_2$ fold change = 0.46). *osm-11* is not changed upon neuronal DAF-2 reduction (Fig. 3b). *osm-11*'s promoter is a direct DAF-16 target²⁹ with both a DBE and a DAE (Supplementary Table 3), indicating that *osm-11* may be a direct target of the hypodermal IIS pathway. *osm-11* encodes a secreted Notch ligand expressed in the hypodermis and spermatheca of adult worms²⁷ (Fig. 3c,d). Finally, OSM-11 is secreted from the

Fig. 3 | Notch signaling components including *osm-11* are required for extended memory. **a**, A total of 15 of 1,436 class I (*daf-2* upregulated) genes are exclusively expressed in the hypodermis and are predicted to encode secreted proteins. **b**, *osm-11* expression is upregulated upon hypodermal but not neuronal DAF-2 degradation. DESeq2 (adjusted *P* values). **c**, *osm-11* is expressed exclusively in the hypodermis. Day 1, *n* = 21 worms. **d**, *osm-11* is not expressed in neurons in young (day 2) and older adults (day 5). *osm-11*::*gfp* (green, hypodermis), neurons = Dll (red). *n* = 15–20 worms/age. **c,d**, Representative images. Scale bars, 100 μ m. **e**, *osm-11* adult-only RNAi knockdown (starting late L4) abolishes memory enhancement induced by hypodermal DAF-2 degradation (STAM on day 3 adults). **f**, Memory evaluated by the rate of change in CI between 0-h and 2-h time points across replicates. **g**, Hypodermis-specific *osm-11* overexpression enhances memory in young (day 2) adults. **h**, Memory evaluated by the rate of change in CI between 0-h and 1-h time points across replicates. **i–k**, *lin-12* knockdown in *flp-18*-expressing neurons (**i**) (RIG, AIY, RIM, and AVA) or the RIG alone (**k**) (*twk-3* promoter) abolishes memory enhancement induced by

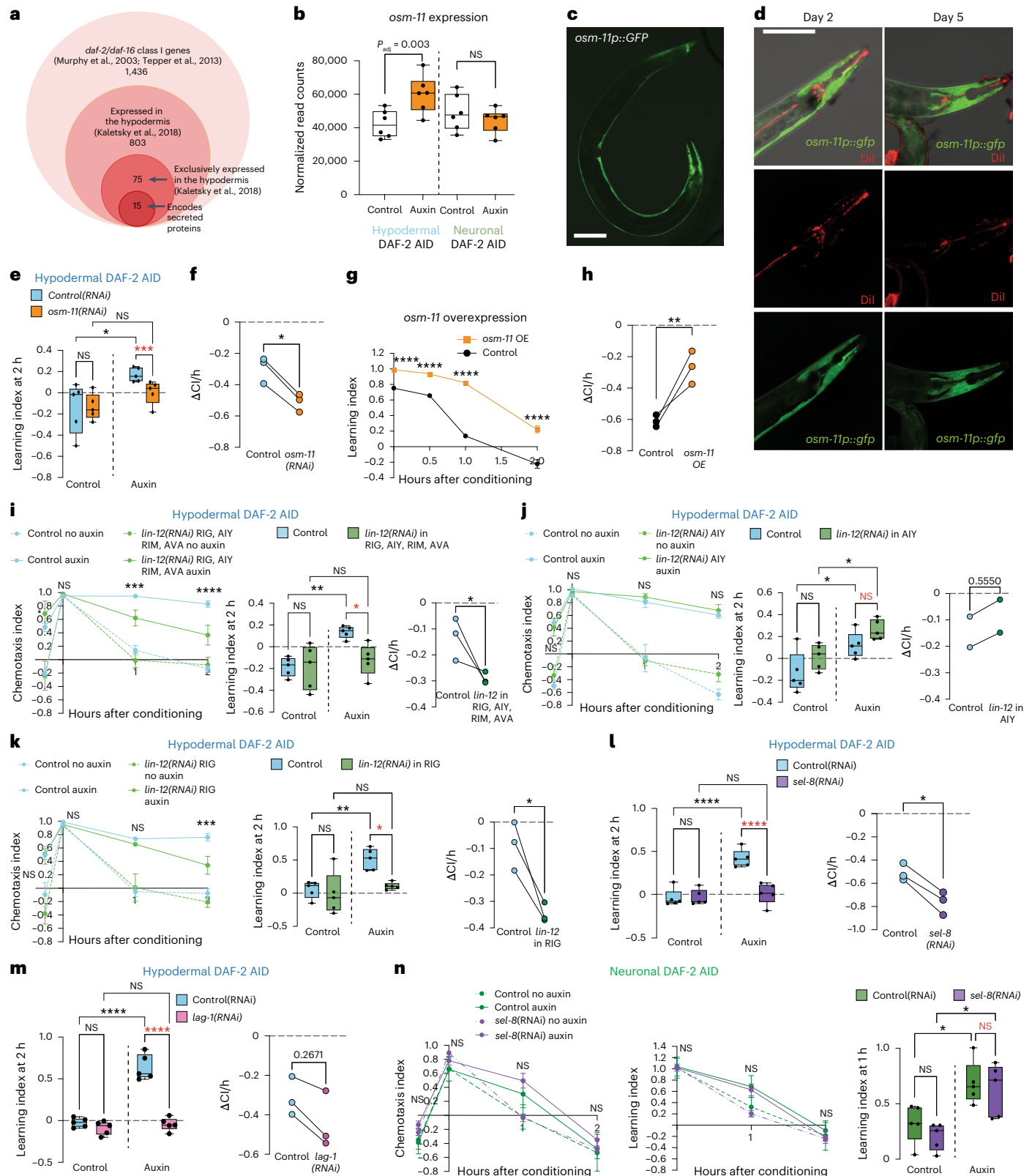
hypodermal DAF-2 degradation, while *lin-12* knockdown in the AIY interneurons does not (**j**). CI plot (left), learning index at 2 h (middle). Memory evaluated as in (**f**). **l,m**, RNAi knockdown of *sel-8* (**l**) or *lag-1* (**m**) (starting late L4) blocks memory enhancement induced by hypodermal DAF-2 degradation (neuronal RNAi-sensitized background). **l**, Memory evaluated as in **h**, or for **m** as in **f**. **n**, *sel-8* adult-only RNAi knockdown is not required for memory enhancement induced by neuronal DAF-2 degradation (neuronal RNAi-sensitized background). *n* = 1 replicate. **e,i–n**, Red indicates comparison of note for genetic interpretation. **b,e,i–n**, Mean $\pm$ s.e.m. Two-way repeated-measures ANOVA and Tukey or Bonferroni post hoc test were used (**g,i–n**). In the box plots, the center line denotes the median, box range indicates the 25th–75th percentile, and whiskers denote minimum–maximum values. *n* $\geq$ 4 chemotaxis plates/time point, each dot represents an individual assay plate with ~100 worms. Representative experiment from two to three biological replicates unless specified. **f,h,i,j–n**, Slope analysis, two-sided unpaired *t*-test. * $P \leq 0.05$, ** $P \leq 0.01$, *** $P \leq 0.001$, **** $P \leq 0.0001$.

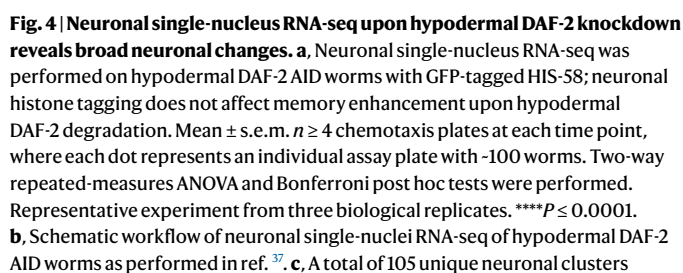
hypodermis and non-autonomously regulates octanol avoidance and quiescence²⁸. OSM-11 is expressed in the seam cells of the hypodermis but not in neurons (*osm-11p::gfp*; Fig. 3c,d). Therefore, the Notch ligand OSM-11 emerged as our top candidate for mediating IIS signaling from the hypodermis to neurons directing memory improvement.

We next tested whether OSM-11 is required for hypodermal-degraded DAF-2 memory extension: *osm-11* knockdown abrogated

the memory increase induced by hypodermal DAF-2 degradation in young adults (Fig. 3e,f and Extended Data Fig. 4a,b), suggesting that reduced hypodermal IIS acts via the secreted protein OSM-11 to regulate associative memory.

To determine if we can extend memory in wild-type worms by increasing Notch signaling, we overexpressed OSM-11 specifically in the hypodermis using the *osm-11* promoter in young adults, and





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tested memory; OSM-11 overexpression increased associative memory (Fig. 3g,h and Extended Data Fig. 4b). Overexpression of *osm-11* slightly increases learning, but also significantly improves memory (Fig. 3g,h). Although OSM-11 overexpression slightly increases lifespan (Extended Data Fig. 4c), its effect on memory is proportionally greater. These results suggest that OSM-11 is required for hypodermal IIS memory enhancement and is sufficient to enhance memory in the absence of additional inputs.

Hypodermal IIS-regulated memory requires the Notch pathway

Notch signaling is activated when Notch ligands bind to the receptors LIN-12 or GLP-1 (refs. 30,31), which are then cleaved, releasing the intracellular signaling transducing fragment, the Notch intracellular domain^{32,33}. The Notch intracellular domain then translocates to the nucleus and binds to the transcription factor LAG-1 and coactivator LAG-3/SEL-8, forming a transcriptionally active ternary complex to regulate gene transcription³⁴. Therefore, we hypothesized that OSM-11 might signal from the hypodermis to non-autonomously activate neuronal Notch signaling. If so, then other key components of Notch signaling might be required in neurons for the memory induced by hypodermal IIS reduction.

To identify candidate-receiving cells for the OSM-11 ligand, we reduced the LIN-12 Notch receptor in a subset of neurons (RIG, AIY, RIM, and AVA interneurons) using a hairpin RNA-mediated interference (RNAi) strain (*flp-18p::lin-12(RNAi)*)^{28,35}. Indeed, knockdown of *lin-12* in those interneurons blocked the memory enhancement induced by hypodermal IIS reduction (Fig. 3i and Extended Data Fig. 4d,e). (*nmr-1p::lin-12(RNAi)* worms were not healthy, preventing analysis of the effect of *lin-12* knockdown in RIM and AVA in memory.) The knockdown of *lin-12* only in AIY neurons (*ttx-3p::lin-12(RNAi)*), an interneuron that is critical for LTAM formation in wild-type worms¹³, did not abrogate memory improvement (Fig. 3j and Extended Data Fig. 4f,g). However, *lin-12* knockdown driven by the RIG-associated *twk-3* promoter³⁶ (*twk-3p::lin-12(RNAi)*) significantly blocked the memory increase induced by hypodermal IIS reduction (Fig. 3k and Extended Data Fig. 4h,i), suggesting that the RIG might be a receiving cell for the OSM-11 ligand through the LIN-12 Notch receptor.

The GLP-1 Notch receptor is also expressed in many neurons, including multiple ciliated sensory neurons²⁸; *glp-1* knockdown in ciliated neurons, even in the absence of auxin treatment, caused severe general learning and chemotaxis defects (Extended Data Fig. 5a–c), preventing the subsequent analysis of memory. To test whether GLP-1 might act in the RIG neuron, we constructed the hairpin RNAi of *glp-1* under the *twk-3* promoter; however, loss of *glp-1* in the RIG had no effect on hypodermal DAF-2 reduction-mediated memory (Extended Data Fig. 5d,e), suggesting that LIN-12 is the sole Notch receptor required in the RIG neuron for this signaling pathway.

While we were unable to obtain healthy strains reducing *sel-8* and *lag-1* specifically in the RIG neuron, knockdown of *sel-8* (Fig. 3l and Extended Data Fig. 5f,g) or *lag-1* (Fig. 3m and Extended Data Fig. 5h,i) in

neuronal RNAi-sensitive, hypodermal DAF-2-degraded adults reduced hypodermal DAF-2 degradation-mediated memory improvement. Knocking down *sel-8* had no effect on memory improvement induced by neuronal DAF-2 degradation (Fig. 3n), suggesting that neuronal DAF-2 degradation increases associative memory through mechanisms distinct from Notch signaling. Together, our results suggest that components of the Notch signaling pathway, including LIN-12 function in the RIG, LAG-1, and SEL-8, are required in neurons for the extended memory induced by hypodermal DAF-2 degradation.

Single-nucleus RNA-seq of hypodermal DAF-2-degred worm neurons

Next, we sought to better understand how hypodermal IIS and Notch signaling might affect transcription in individual neurons. Previously, we carried out adult pan-neuronal RNA-seq of wild-type and *daf-2* worms to identify neuronal genes that are required for *daf-2*'s extended memory¹⁰. We also recently identified transcriptional changes in individual neurons altered in *daf-2* adults³⁷ using a method we recently adapted from mammalian neuronal single-nucleus RNA-seq protocols³⁷; single-nucleus RNA-seq allows transcriptional analysis of individual neurons. To identify transcriptional changes in individual neurons altered by hypodermal DAF-2 degradation, we first constructed a hypodermal DAF-2-AID strain expressing a pan-neuronal GFP-histone tag; these worms retain the ability to enhance memory after auxin treatment (Fig. 4a). We isolated GFP neuronal nuclei from day 2 no-auxin control and hypodermal DAF-2-degraded (+ auxin) worms and performed single-nucleus RNA-seq (Fig. 4b). We obtained 122,834 neuronal nuclei with an average of 456 unique molecular identifiers per cell and detected 370 genes per cell (Extended Data Fig. 6). We successfully identified 106 clusters that match 118 specific neuron classes (Fig. 4c), with no missing or extra clusters after auxin treatment (Supplementary Table 4).

Gene Ontology analysis of neuronal genes differentially expressed upon hypodermal DAF-2 degradation revealed upregulation of pathways related to voltage-gated channel activity, calcium signaling, G-protein-coupled receptor activity, and synaptic vesicles (Fig. 4d,e). Additionally, MAPK signaling, which is known to be required for synaptic plasticity and memory³⁸, was enriched among the neuronal upregulated genes. These data suggest that the reduction of hypodermal DAF-2 may enhance memory by increasing neuronal activity and synaptic plasticity^{39–41}. The FOXO and mTOR longevity-related pathways were enriched among neuronal genes (Fig. 4e), consistent with the role of IIS in *C. elegans*^{15,16,23,42,43}. Ribosomal genes were downregulated in neurons (Extended Data Table 2), implying that there might be reduced ribosomal biogenesis in neurons following hypodermal IIS reduction, which may conserve energy and promote neuronal health^{44,45}.

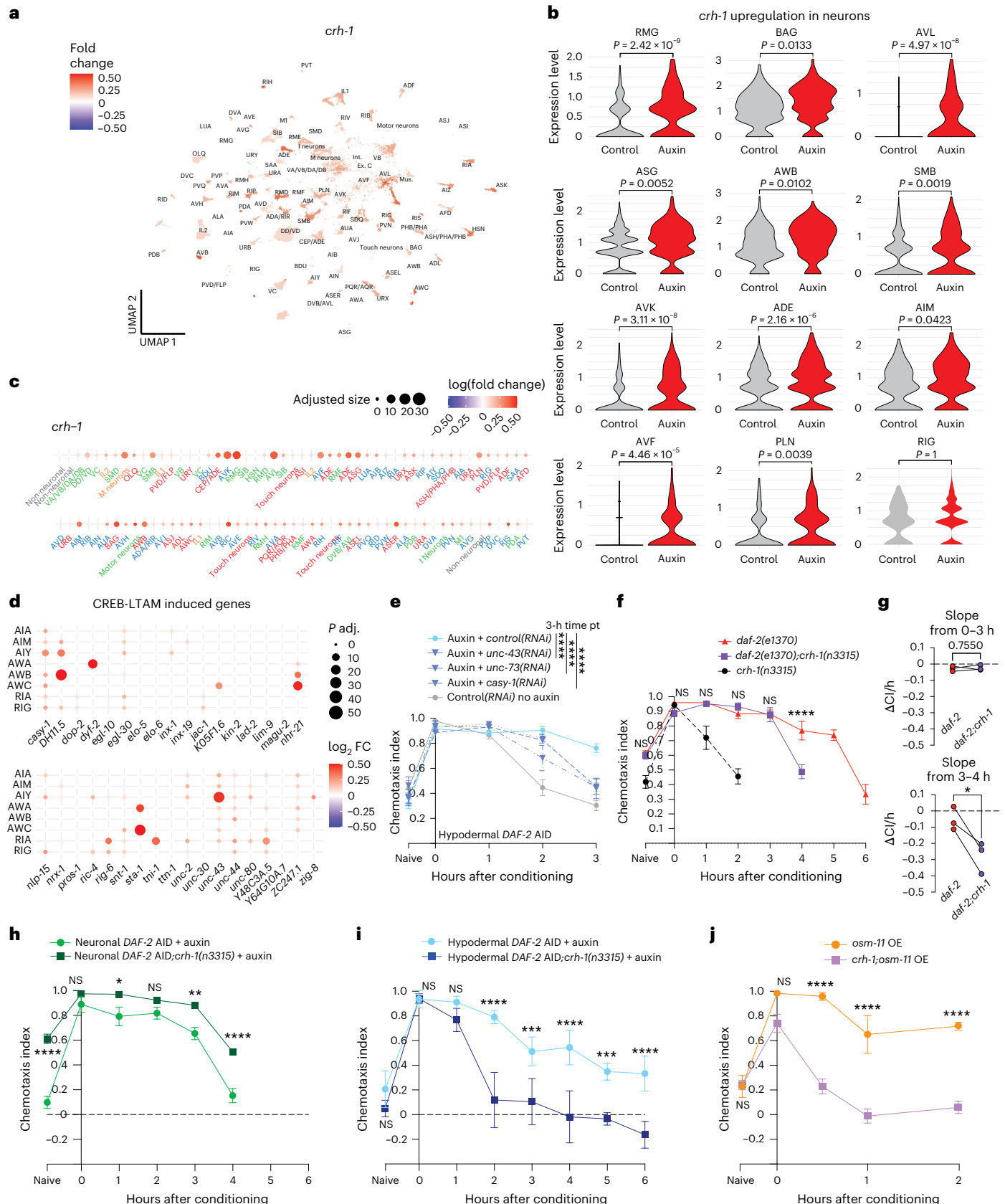
We were next interested in neuron-specific changes that might mediate enhanced memory upon hypodermal IIS reduction. The AWC pair of chemosensory neurons detect a variety of chemoattractants, including butanone, and are required for short-term associative learning and memory in both wild-type and *daf-2* animals^{3,13,37,46,47}.

Fig. 5 | CREB is required for hypodermal IIS reduction-induced memory, but not for neuronal IIS reduction-induced memory. a, *crh-1* is differentially expressed in many neurons upon hypodermal DAF-2-degradation. Red indicates upregulated in +auxin condition compared to no-auxin control. **b**, *crh-1* levels in neurons with significantly upregulated expression upon hypodermal DAF-2 reduction. Wilcoxon rank-sum test. FDR ≤ 5%. **c**, *crh-1* expression is increased broadly across the nervous system. Fold change across all neurons is shown. **d**, Hypodermal DAF-2(–)-upregulated CREB/LTAM genes in select neurons⁷. Wilcoxon rank-sum test. **e**, Adult-only RNAi knockdown (starting from late L4) of the known CREB/LTAM genes *unc-43*, *unc-73*, or *casy-1* impairs 3-h associative memory extension hypodermal DAF-2(–) animals. *n* ≥ 4 chemotaxis plates at each time point, where each dot represents an individual assay plate with ~100 worms. Hypodermal DAF-2 AID performed in neuronal RNAi-sensitized background.

f, Young (day 1) *daf-2(e1370)* mutants have extended memory beyond 5 h after a single training session, while *daf-2(e1370);crh-1(n3315)* worms' memory lasts 2 h less. *crh-1(n3315)* worms' memory lasts <2 h. **g**, Memory was evaluated by calculating the rate of change in CI between the learning-indicated time points across replicates. Two-sided unpaired *t*-test. **h**, Loss of CREB does not affect the memory extension of neuronal DAF-2 degradation until hour 4 after a single training session in young (day 2) adult animals. **i**, *crh-1* mutation abolishes the memory induced by hypodermal IIS reduction. **j**, *crh-1* is required for learning and memory in *osm-11*-overexpressing worms. **e,f,h–j**, Representative experiment from three biological replicates. Two-way repeated-measures ANOVA and Bonferroni post hoc tests were performed. Mean ± s.e.m. *n* ≥ 4. **P* ≤ 0.05, ***P* ≤ 0.01, ****P* ≤ 0.001, *****P* ≤ 0.0001.

We previously performed single-nucleus RNA-seq on wild-type *daf-2* mutant animals and identified transcriptional changes in the AWC neurons that result from whole-worm IIS reduction³⁷. We compared the hypodermal IIS and whole-worm IIS changes in the AWC neurons

and found that DAF-2 reduction exclusively in the hypodermis accounts for ~20% (7/36) of the *daf-2* AWC-upregulated genes (Fig. 4f). Shared upregulated genes include neprilysin (*nep-26*), *ist-1*, *C36C9.5*, Rho GEF Trio/*unc-73*, the VHI phosphatase *vhp-1*, and the transcription factors



STAT/*sta-1* and *nhr-21* (Fig. 4g). Notably, *nep-26*, *vhp-1*, and *unc-73* have previously been shown to be required for the extended short-term memory of *daf-2* mutants³⁷.

Broad CREB upregulation in hypodermal DAF-2 AID worm neurons

The transcription factor CREB and its downstream transcriptional targets are essential for long-term memory^{3,5,7}. Our single-nucleus RNA-seq results showed that *crh-1*/CREB was upregulated in many neurons upon DAF-2 degradation in the hypodermis (Fig. 5a–c); notably, RIG is not the major site of CREB expression change, suggesting that CREB may be an indirect rather than direct target of Notch signaling. We identified 35 neuronal genes upregulated by hypodermal DAF-2 depletion that are also CREB targets induced by long-term memory training (CREB/LTAM⁷; Fig. 5d), and 36 neuronal genes induced upon hypodermal DAF-2 degradation that are orthologs of genes upregulated by gain-of-function GNAQ expression in the mammalian hippocampus (Extended Data Fig. 6i). Most of these overlapping genes, including those that we previously determined to be crucial for learning and memory (for example, *casy-1*, *egl-30*, *irk-3*, *kin-2*, *pkc-1*, *unc-43*, and *unc-73*), exhibited increased expression in memory-related and sensory-related neurons—AIY, AIM, RIA, AIA, AWA, AWB, and AWC (Fig. 5d and Extended Data Fig. 6i). For instance, *casy-1*, which encodes the transmembrane protein calyntenin, is upregulated across all three conditions (hypodermal DAF-2 degradation, CREB-LTAM and GNAQ(gf)) and is required for learning^{3,7,12}. Similarly, *unc-43*, the *C. elegans* ortholog of the CaMKII type II calcium/calmodulin-dependent protein kinase, is required for memory maintenance^{7,8} and is induced by all three treatments.

We next investigated whether these induced genes are required for the memory extended by hypodermal IIS signaling. We used hypodermal DAF-2 AID worms with a neuronal RNAi-sensitive background to knock down specific genes and then tested memory duration. We focused on genes that are also regulated by CREB-LTAM and/or GNAQ(gf), or are known to be involved in learning and LTAM (*unc-43*, *unc-73*, and *casy-1*)^{3,8,12,13}. RNAi treatment was initiated in adult animals to avoid causing developmental defects, and no motility defects were observed (Extended Data Fig. 7a). We found that all three genes are required for extended memory after hypodermal DAF-2 degradation (Fig. 5e and Extended Data Fig. 7b–f). Together, our data indicate that reduced hypodermal IIS bolsters memory maintenance via upregulation of CREB and multiple target genes in neurons.

CREB is required only for hypodermal IIS-dependent memory

We previously found that CREB is required for long-term memory, and increasing CREB activity extends long-term memory, as CREB is activated during spaced training^{3,7}; however, a role for CREB in extending short-term memory after a single training session is less well studied. The upregulation of *crh-1* and CREB/LTAM target expression in many neurons (Fig. 5d) suggested that CREB may contribute to the enhanced memory resulting from hypodermal DAF-2 degradation, and may at least partially contribute to the prolonged short-term memory exhibited by *daf-2* mutants (Fig. 1a and Extended Data Fig. 1d,e).

To investigate whether CREB is required for *daf-2*'s extended short-term memory, we tested the duration of memory after 1 h of training in *daf-2(e1370);crh-1(n3315)* double mutants. While *daf-2(e1370)* mutants' memory lasts up to 6 h, loss of CREB (*daf-2(e1370);crh-1(n3315)* double mutants) reduces memory to 4 h (Fig. 5f–g and Extended Data Fig. 7g–i), which suggests that CREB is partially required for the full extended memory of *daf-2* mutants.

Since DAF-2 degradation in the hypodermis and neurons, not other tissues, induces memory enhancement, we then tested where the partial CREB contribution in *daf-2* mutants' memory originates from. Loss of CREB increases naive CI, obscuring learning (Extended Data Fig. 7h) but has no effect on neuronal DAF-2 degradation-induced memory (Fig. 5h and Extended Data Fig. 8a–d), while loss of CREB from

hypodermal DAF-degraded worms completely abrogates memory from 2 h onward (Fig. 5i and Extended Data Fig. 8e–h), suggesting that hypodermal IIS acts through CREB to extend memory, but neuronal IIS does not. Loss of *crh-1* decreased the improved learning and memory induced by *osm-11* overexpression, as well (Fig. 5j and Extended Data Fig. 8i,j), further suggesting that the memory increase exhibited by OSM-11 overexpression is similarly dependent on CREB. Together, these data suggest that the CREB-independent portion of *daf-2*'s extended memory (~2 h) is due to its direct DAF-2 reduction in neurons, while hypodermal IIS regulation of CREB via OSM-11 accounts for the remaining fraction of *daf-2*'s extended memory. We found that CREB is not required in the RIG neuron for the extended memory of hypodermal DAF-2 degradation animals (Extended Data Fig. 8k–m), suggesting that Notch signaling in the RIG must communicate to additional neurons, rather than acting in RIG itself, to regulate CREB.

Reduced hypodermal IIS in old worms slows cognitive aging

We previously found that aged worms lose the ability to learn and to remember with age, but reduction of insulin signaling results in prolonged maintenance of learning and associative memory³. However, these experiments were performed using mutants in which IIS was reduced in all tissues^{3,10}. We wondered whether we could ameliorate aging-induced memory loss by intervening later in life, particularly when animals are already experiencing pronounced memory impairment. (By day 4–5 of adulthood, wild-type worms have no LTAM but can still move (Extended Data Fig. 9a–d); by day 7–8, learning and short-term memory are also abrogated³.) To address this question, we degraded DAF-2 in the hypodermis starting in mid-adulthood, when cognitive function is diminishing in wild-type animals³, and tested the worms' learning and memory. Because memory declines earlier in adulthood than learning³, we started auxin treatment on day 5 and performed memory assays on day 6, when *daf-2* mutants are still capable of learning and memory, but wild-type worms are not (Extended Data Fig. 9)²⁶. Remarkably, DAF-2 degradation specifically in the hypodermis on day 5 significantly increased associative memory in day 6 worms, an age when wild-type worms can learn but have no memory (Fig. 6a and Extended Data Fig. 10a,b), indicating beneficial effects from the reduction of hypodermal IIS in cognitive aging. By contrast, DAF-2 degradation specifically in the neurons on day 5 did not increase associative memory in day 6 worms (Fig. 6b and Extended Data Fig. 10c,d). While both hypodermal and neuronal DAF-2 degradation in young (day 2) worms significantly improves 1-h learning index, only hypodermal DAF-2 degradation improves 1-h learning index in aged (day 6) animals (Fig. 6c and Extended Data Fig. 10e). Notably, the magnitude of memory increase from hypodermal DAF-2 degradation in midlife is comparable to the memory improvement observed in young worms (Fig. 6c), indicating the potential of this midlife intervention to mitigate cognitive aging. Together, these results suggest that reducing hypodermal insulin signaling via DAF-2 degradation specifically in the hypodermis, even in midlife, can enhance associative memory. This difference between hypodermal and neuronal DAF-2 degradation in aged worms highlights the difference between molecular mechanisms involved in the two pathways of IIS-mediated memory extension.

Our previous whole-worm RNA-seq study of aging animals²⁶ revealed that *osm-11* expression decreases with age (Fig. 6d). Worms overexpressing OSM-11 tested at day 5 of adulthood—an age when wild-type worms have poor learning ability and no functional memory³ (see control)—showed both significantly enhanced learning and memory (Fig. 6e–g and Extended Data Fig. 10f). Our data suggest that OSM-11 overexpression even in aged animals is sufficient to confer learning and memory improvements, consistent with our data on hypodermal DAF-2 degradation on older animals.

We previously found that aged *daf-2* mutants retain learning and memory longer than do wild-type worms²⁶. To determine if

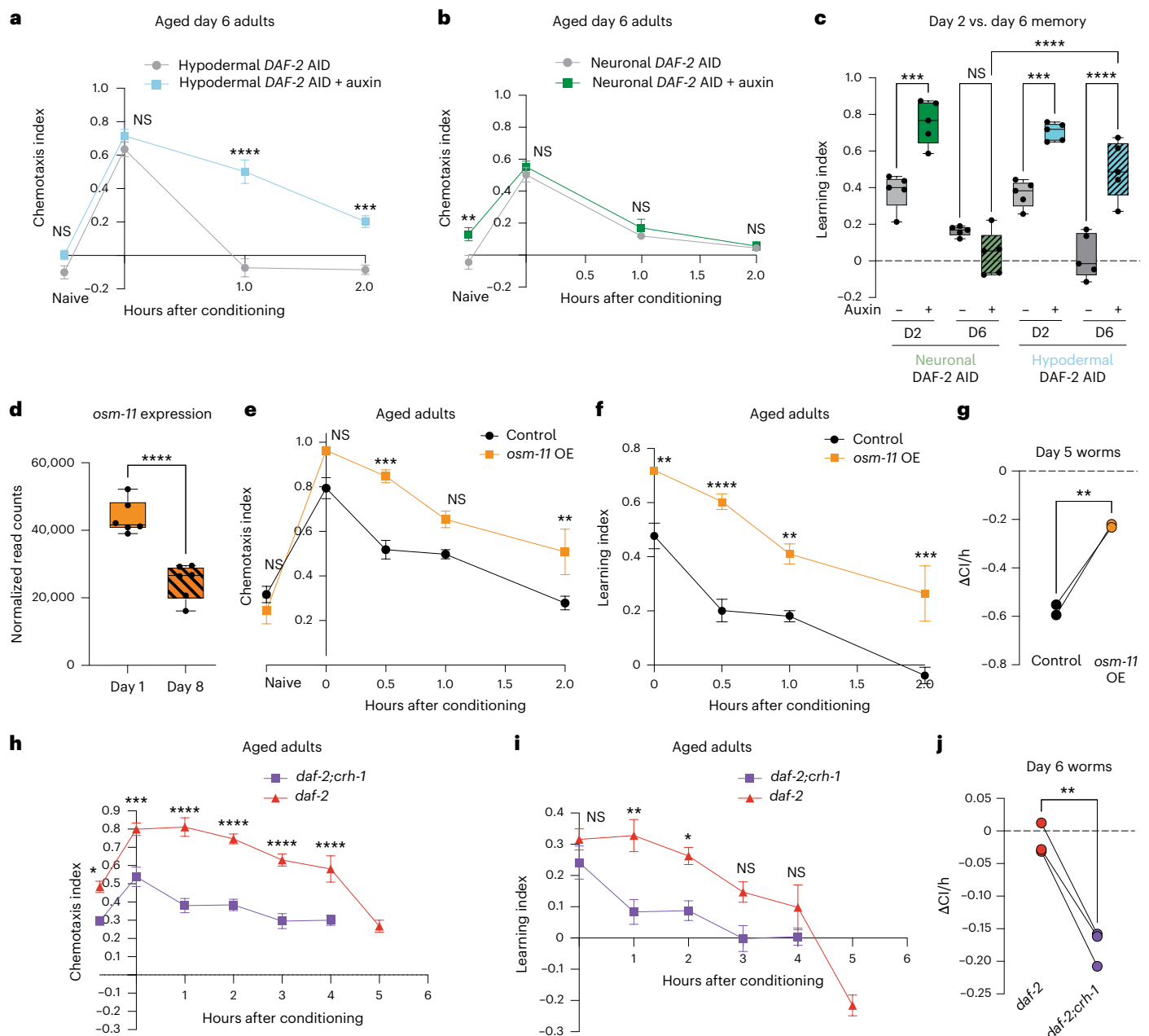


Fig. 6 | Hypodermal DAF-2 degradation in aging worms slows cognitive decline through CREB-dependent pathway via Notch ligand OSM-11. a, Hypodermis-specific DAF-2 degradation in old adult animals improves STAM. **b**, Neuron-specific DAF-2 degradation in old adult animals does not increase STAM. **a, b**, Auxin started on day 5, while STAM was performed on day 6. **c**, Comparison of 1-h memory in young (day 2) versus old (day 6) adult animals with hypodermis-specific or neuron-specific DAF-2 degradation. Only hypodermis-specific DAF-2 degradation improves old (day 6) adult memory. **d**, *osm-11* expression decreases with age²⁶. **c, d**, In the box plots, the center line denotes the median, box range indicates the 25th–75th percentile, and whiskers denote minimum–maximum values. **e, f**, OSM-11 overexpression increases memory in aged (day 5) adults. CI (e)

and learning index (f) are shown. **g**, Memory was evaluated by calculating the rate of change in CI between the learning and 0.5-h time points across replicates. **h, i**, Aged (day 6) *daf-2(e1370);crh-1(n3315)* learning and STAM is impaired relative to *daf-2* worms. CI (h) and learning index (i) are shown. **j**, Slope analysis as in g using learning and 1-h time points. **a, b, e, f, h, i**, Representative experiment from three biological replicates. Mean $\pm$ s.e.m. Two-way repeated-measures ANOVA using data. Bonferroni post hoc tests. $n \geq 4$ chemotaxis plates at each time point, where each dot represents an individual assay plate with ~80 worms. Representative experiment for three biological replicates. * $P \leq 0.05$, ** $P \leq 0.01$, *** $P \leq 0.001$, **** $P \leq 0.0001$. **c, d, g, j**, Two-sided unpaired *t*-test.

the memory improvement of older *daf-2* mutants is the result of an integrated effect of CREB-dependent memory signaling from hypodermal IIS and CREB-independent extended STAM signaling from neuronal IIS, we tested memory duration in *daf-2;crh-1* mutants on day 6, an age at which *daf-2* mutants still have memory but wild-type worms do not²⁶. While day 6 *daf-2* mutants are still capable of memory beyond 3 h after conditioning, *daf-2;crh-1* mutants have

no learning or memory ability, suggesting that *crh-1* is largely responsible for *daf-2*'s memory extension at mid-adulthood (Fig. 6h–j and Extended Data Fig. 10g). Together, our results indicate that, unlike the combined effects from hypodermal and neuronal IIS signaling in young *daf-2* mutants, the enhanced memory of middle-aged *daf-2* mutants is fully dependent on hypodermal IIS signaling and CREB activity.

Discussion

In this study, we present evidence of non-autonomous regulation of CREB-dependent associative memory mediated by the insulin pathway specifically in the hypodermis of *C. elegans* via Notch signaling from the hypodermis to neurons. We had previously identified transcriptional changes in memory⁷ and expression changes specifically in neurons in *daf-2* animals¹⁰, but possible contributions to memory function from *C. elegans*' non-neuronal tissues were unknown. Therefore, we were surprised to discover that degradation of the DAF-2 receptor solely in the worm's hypodermis—a large syncytial tissue that we previously found to have a liver-like metabolic transcriptional signal²⁰—increases memory to a greater magnitude and duration than degradation of DAF-2 in neurons.

C. elegans with elevated insulin signaling—that is, the opposite direction of *daf-2* mutants or DAF-2 degradation—have learning impairments associated with high neuronal levels of kynurenic acid, whose precursor is generated in epidermal cells^{48,49}, indicating negative involvement of non-neuronal tissues in learning and memory; however, the fact that reduced insulin signaling in non-neuronal tissues could have beneficial effects on memory had not been previously reported. In addition, we previously found that memory was extended beyond 5 h after one training session in *daf-2* mutants, but whether any part of that memory is dependent on CREB was previously unknown. Here, we found that DAF-2 degradation exclusively in the hypodermis extends memory even longer than the memory extension induced by neuronal DAF-2 degradation, primarily through CREB activity.

Our single-nucleus sequencing data and subsequent memory assays implicated CREB in the regulation of hypodermal IIS-regulated memory. While memory extension by neuronal DAF-2 depletion is independent of CREB, hypodermal DAF-2 degradation requires CREB activity, indicating that the extended memory exhibited by *daf-2* mutants is the result of the combined effect of extended STAM originating from neuronal IIS reduction and the expression of CREB and its targets in hypodermal IIS reduction neurons before training. By contrast with CREB-dependent long-term memory, which forms upon increased CREB activity during spaced training^{3,7}, the improvements in learning and memory we observe in animals with hypodermal DAF-2 degradation are due to increased basal CREB activity and neuronal transcriptional changes prior to training. Furthermore, our transcriptomic and memory assay results suggest that the upregulation of a diffusible Notch ligand, OSM-11, mediates this memory improvement via downstream components of the Notch signaling pathway in neurons, including the Notch receptor LIN-12, the transcription factor LAG-1, and the coactivator SEL-8. Thus, the memory extension exhibited by *daf-2* mutants is the result of the integration of enhanced short-term memory regulated by neuronal IIS and the formation of CREB-dependent memory induced by hypodermal IIS regulation of Notch signaling (Extended Data Fig. 10h). Surprisingly, *daf-2*'s improvement in learning and memory with age relative to wild type is entirely due to the hypodermal IIS–Notch–CREB activity, rather than IIS signaling in neurons. Together, these results define a mechanism for the regulation of memory in *C. elegans* by a non-neuronal tissue, the hypodermis.

Notch signaling is an evolutionarily conserved pathway, and its roles in development, cancer, and nervous system development in mammals^{50,51}, as well as its role in *C. elegans* development, have been studied extensively^{52–54}. However, Notch signaling's post-developmental adult roles in *C. elegans* are limited^{27,28}. Previous studies uncovered the neuron-autonomous role of neuronal Notch signaling in memory regulation in flies and mice: knocking down the Notch receptor specifically in neurons of the mushroom body, the brain center that plays an essential role in *Drosophila* learning and memory, causes deficits in long-term memory^{55,56}, and postnatal disruption of Notch signaling in the mouse brain causes defects in learning and memory⁵⁷. Additionally, Notch signaling in the brain regulates mammalian hippocampal plasticity through CREB signaling⁵⁸ also through

cell-autonomous signaling. In *C. elegans*, Komatsu et al. showed that OSM-11 is a secreted Notch ligand expressed in the hypodermis²⁷, and Singh et al. found that OSM-11 is secreted from the hypodermis and acts non-autonomously to regulate *C. elegans* octanol avoidance and quiescence²⁸, suggesting that Notch signaling regulates some neuronal functions by receiving signals from peripheral tissues. However, Notch signaling's regulation of neuronal CREB activity and its role in memory, as well as its regulation by IIS, was previously unknown; while IIS is known to regulate direct effectors of longevity and other functions through the activity of the DAF-16 transcription factor¹⁵, its impact on the Notch signaling pathway in *C. elegans* neurons has not been previously reported. Together with the previous work on Notch signaling in worms, flies, and mice^{27,28,55,57,58}, our data indicate that the IIS–Notch–CREB axis of memory regulation may be conserved.

The *C. elegans* hypodermis is one of the worm's largest metabolic tissues and shares transcriptional similarity with human liver and blood plasma²⁰. The hypodermis also aids in adaptation to external environmental changes, such as food scarcity and changes in osmolarity⁵⁹. Our findings suggest that it is possible that this metabolic tissue may aid in remembering food odors: the hypodermis' non-autonomous memory regulation might allow a long-lasting mechanism to respond more readily to future environmental changes, with different temporal dynamics than regulation in the neurons. Our observation of non-autonomous regulation of *C. elegans* neuronal function, and specifically learning and memory, suggests that the animal might have redundant systems in place to modulate memory function, a system that might be helpful under conditions of sustained food limitation. By engaging signaling in the neurons downstream of Notch signaling, worms may be able to have a more sustained memory response than insulin signaling in neurons alone might achieve, particularly in aging animals.

Interestingly, there may be a parallel in the effect of the mammalian liver on memory: the administration of systemic blood plasma from exercised aged mice—which secrete factors from the liver into the blood—improves cognitive function and neurogenesis in sedentary aged recipient mice⁶⁰. Likewise, Alzheimer's disease model rats receiving administration of a potent metabolic regulator, fibroblast growth factor 21, which is mainly produced by the liver, also exhibit ameliorated neurodegeneration and enhanced learning and memory⁶¹. Together, these data reveal a critical connection between non-neuronal metabolism and memory regulation, suggesting a similar mechanism across different animal models, and providing therapeutic insight to ameliorate cognitive aging or to treat neurodegenerative diseases by regulating metabolism.

Limitations of the study

Our genetic data support a role for LIN-12 function in the RIG neuron in Notch signaling regulation of memory, but we were unable to test the role of *sel-8* or *lag-1* specifically in the RIG; moreover, our single-nucleus RNA-seq data suggest that upregulation of *crh-1*/CREB expression in many neurons as a result of OSM-11/Notch signaling is widespread, rather than solely acting through the RIG. The most logical conclusion is that this regulation may occur indirectly via neuropeptide signaling⁶², of which there are hundreds of pair possibilities.

Methods

General worm maintenance

All strains were cultured using standard methods⁶³. For all experiments, worms were maintained at 20 °C on plates made from nematode growth medium (NGM: 3 g l⁻¹ NaCl, 2.5 g l⁻¹ Bacto-peptone, 17 g l⁻¹ Bacto-agar in distilled water, with 1 ml l⁻¹ cholesterol (5 mg ml⁻¹ in ethanol), 1 ml l⁻¹ 1 M CaCl₂, 1 ml l⁻¹ 1 M MgSO₄ and 25 ml l⁻¹ 1 M potassium phosphate buffer (pH 6.0) added to molten agar after autoclaving) or high growth medium (HGM: NGM recipe modified as follows: 20 g l⁻¹ Bacto-peptone, 30 g l⁻¹ Bacto-agar and 4 ml l⁻¹ cholesterol (5 mg ml⁻¹ in ethanol); all other components same as NGM), with OP50 *Escherichia coli* for

ad libitum feeding. To synchronize experimental animals, eggs were collected from gravid hermaphrodites by exposing the animals to an alkaline-bleach solution (for example, 1.5 ml sodium hypochlorite, 0.5 ml 5 N KOH, 8.0 ml water), followed by repeated washing of collected eggs in M9 buffer (6 g l⁻¹ Na₂HPO₄, 3 g l⁻¹ KH₂PO₄, 5 g l⁻¹ NaCl and 1 ml l⁻¹ M MgSO₄ in distilled water).

C. elegans strains

Strains are listed in Supplementary Table 5.

Primers used in this study

Three promoters were used to drive the sense and antisense expression of target genes: *Pttx-3* (AIY interneurons), *Ptwk-3* (RIG interneurons) and *Pnmr-1* (RIM and AVA interneurons).

ttx-3 promoter⁶⁴:

ttx-3 Pf = tcattgcatattcgattttttcagtaaagc

ttx-3 Pr = ttgtacaccgaagacaattattatg

twk-3 promoter³⁶:

twk-3 Pf = tacgccatcaattagattttttatgtctg

twk-3 Pr = tttaagaagaacaaggaaaaagttgaag

nmr-1 promoter⁶⁵:

nmr-1 Pf = gatgattatggaacaaactcagaatttaag

nmr-1 Pr = atctgtaacaaaactaaagttgtcgtg

lin-12 for sense and antisense constructs:

Pf = atgcggatccctacgatttg

Pr = acactagccccttgctgaatc

lag-1 for sense and antisense constructs:

Pf = ggatgtggaatggaaattggagg

Pr = ctcttccatgcaactcgacacgc

glp-1 for sense and antisense constructs:

Pf = gtctcgactgcaaaactcctctattttc

Pr = acaaatgttgctcgagttcaatccatg

crh-1 for sense and antisense constructs:

Pf = atggccacaatggcgagcacctc

Pr = tcacattccgctcttttctttcggc

PCR products of *twk-3p*- or *ttx-3*-driven sense and antisense constructs were injected at 100 ng μl⁻¹ each with 2 ng μl⁻¹ *myo-2p::GFP* marker.

CQ801 was generated by injection at 10 ng μl⁻¹ of a plasmid containing the genomic coding sequence of *osm-11* gene with the *unc-54* 3' untranslated region expressed under the control of 2.5 kb of upstream genomic DNA. As a co-injection marker and reporter of *osm-11* expression, the same *osm-11* promoter was used to express GFP with an *unc-54* 3' untranslated region.

Auxin treatment

For auxin experiments, the standard HGM molten agar was supplemented with 2.5 ml 400 mM indole-3-acetic acid per 1 l of HGM: Alfa Aesar (A10556) freshly prepared in ethanol and plates were seeded with *E. coli* for ad libitum feeding. Synchronized AID worms were transferred to HGM with auxin plates for 16–20 h before behavior assay. For midlife experiments, worms were transferred at the L4 larval stage onto HGM plates supplemented with 500 ml l⁻¹ 0.1 M 5-Fluoro-2'-deoxyuridine (FUDR) for a final concentration of 0.05 M FUDR and were transferred back to standard HGM with or without auxin 20 h before memory assay.

RNAi treatment

For RNAi experiments, the standard HGM molten agar was supplemented with 1 ml l⁻¹ 1 M isopropyl β-D-1-thiogalactopyranoside and 1 ml l⁻¹ 100 mg ml⁻¹ carbenicillin, and plates were seeded with HT115 *E. coli* for ad libitum feeding. All RNAi treatment starts from L4 larval stage and lasts for 3 days. RNAi experiments were performed using the standard feeding RNAi method. Bacterial clones expressing the control construct (empty vector, pL4440). All RNAi clones were sequenced before use.

Pavlovian appetitive associative assay

Animals were trained and tested for short-term memory as previously described³. Briefly, synchronized young or aged adult hermaphrodites were washed from HGM, RNAi or HGM supplemented with auxin plates with M9 buffer, allowed to settle by gravity, and repeatedly washed three more times with M9 buffer. Then animals were starved for 1 h in M9 buffer. For conditioning (food and 10% 2-butanone pairing), worms were then transferred to 10-cm NGM conditioning plates (seeded with OP50 *E. coli* bacteria and with 18 ml 10% 2-butanone (Acros Organics) dissolved in ethanol on the lid) for 1 h at 20 °C. After conditioning, the trained worms were tested for chemotaxis toward 10% butanone versus an ethanol control either immediately (0 h) or after being transferred to 10-cm NGM holding plates with fresh OP50 *E. coli* bacteria for specified time intervals (30 min–2 h). Before spotting 10% butanone or ethanol control on assay plates, 10% sodium azide (diluted using two parts sodium azide and one part nuclease-free water) was added to paralyze worms for analysis of the populations arriving at each odorant spot.

Chemotaxis indices were calculated as follows:

$$\text{Chemotaxis index} = \frac{\text{no. of worms Butanone} - \text{no. of worms Ethanol}}{\text{no. of worms Total} - \text{no. of worms Origin}}$$

The calculation for the learning index is:

$$\text{Learning index} = \text{Chemotaxis Index Trained} - \text{Chemotaxis Index Naive}$$

Learning indices for extrachromosomal transgenic strains were analyzed by hand counting GFP-positive and GFP-negative worms at different locations at individual time points on the chemotaxis plates. Control animals for these experiments were the transgenic worms' GFP-negative siblings.

Chemotaxis assay

Synchronized adult worms were tested for chemotaxis to 1% benzaldehyde in ethanol or 10% pyrazine in ethanol, using standard, previously described chemotaxis assay conditions⁴⁶.

Microscopy

For Dil, worms were grown on regular seeded HGMs until day 2 adults (for young worms Dil staining), or until L4 and transferred to HGMs supplemented with FUDR to day 4 and then transferred to regular HGMs until day 5 (for older worms Dil staining). On the day of imaging, well-fed worms were washed three times by M9, and then resuspended in 1 ml M9 with 5 μl Dil stock solution (2 mg ml⁻¹ Dil (Molecular Probes, D-282) in dimethyl formamide) for 3 h on a slow shaker at room temperature. Worms were washed with M9 twice before transferring them onto agar pads with sodium azide to visualize by Nikon Eclipse Ti microscope. z-stack multi-channel (DIC, mCherry/RFP and GFP) images of day 2 or 5 adult worms were acquired at ×60 magnification. Images were analyzed using Nikon NIS elements software (v5.42.03). For Auxin imaging experiments, worms were treated with auxin per the auxin treatment method.

Worm FACS (population enrichment)

Array-carrying adult D1 worms were sorted using COPAS Large Particle Sorter to select for fluorescent pharynx (array-carrying worms). This enriched population was bleached to synchronize eggs, and once grown, D1 animals were subsequently moved to auxin for STAM experiment, where fluorescent worms were counted manually.

RNA isolation

Synchronized day 2 adult worms were collected in M9 and washed repeatedly 3 times to remove excess bacteria. Worm pellets were crushed in liquid nitrogen and transferred to 850 μl TRIzol LS. Total

RNA was extracted using the standard TRIzol/chloroform/isopropanol method followed by DNase digestion and Qiagen RNeasy Mini kit. Agilent Bioanalyzer RNA Pico chips were used to assess the quality and quantity of isolated RNA. mRNA libraries were prepared using the RNA-seq directional library prep on the Apollo 324 robot and were sequenced (65-nucleotide paired-end) on the Illumina NovaSeq S1 100-nucleotide flowcell v1.5 platform (yields ~1.3–1.6 billion reads).

RNA-seq data analysis

RNA-seq analysis was performed as previously described. FASTQC was used to assess read quality scores. The universal Illumina adaptor sequences were trimmed using Cutadapt v1.6⁶⁶. The trimmed reads were mapped to the *C. elegans* genome (UCSC Feb 2013, ce11/ws245) using STAR⁶⁷. The reads aligning to individual genes were counted using htseq-counts (mode: union), and DESeq2 was used for differential expression analysis. Genes with an adjusted *P* value ≤ 0.05 were considered significantly differentially expressed.

Tissue query

worm.princeton.edu was used for tissue query from upregulated or downregulated gene lists (DESeq2 genes FDR < 1%).

Secreted protein prediction

SignalP-5.0 (DTU Health Tech) was used for secreted protein prediction from 75 genes exclusively expressed in the hypodermis.

Neuronal nuclei isolation

C. elegans neuronal nuclei were isolated using the following methods³⁷, modified from the mammalian single-nucleus RNA-seq protocol from 10x Genomics. Worms (CQ756) grew on regular HG until day 1 and were transferred to HG or HG + auxin plates on day 1. Then, ~400 μ l of day 2 worms CQ756 were washed from HG or HG + auxin plates. Samples were washed 3 $\times$ in M9 buffer and worm pellets were kept on ice until douncing. The 400 μ l worm pellet was transferred to a 1-ml dounce homogenizer (Kimble Glass Tissue Homogenizer, 88-mm overall length, Dounce 1 ml working capacity; 885300-0001) filled with 300 μ l of NP40 lysis buffer (10 mM Tris-HCl (Sigma-Aldrich, T2194; pH 7.4), 10 mM NaCl (Sigma-Aldrich, 59222 C; 5 M), 3 mM MgCl₂ (Sigma-Aldrich, M1028; 1 M), 0.05% Nonidet P40 Substitute (Sigma-Aldrich, 74385), 1 mM dithiothreitol (Sigma-Aldrich, 646563), 1 U μ l⁻¹ RNase inhibitor (Sigma Protector RNase inhibitor; Sigma-Aldrich, 3335402001), Nuclease-free water). Each sample was Dounce homogenized 25–35 $\times$, using only the tight pestle. After 15 strokes, the degree of lysis was examined every five strokes, until appropriate lysis was achieved (no intact worms, with small worm fragments and pieces of empty cuticle present). Samples were moved to 2 ml low-binding microcentrifuge tubes and Dounce homogenizers were rinsed with 1 ml of NP40 lysis buffer; this rinse was added to the tubes. Samples were then incubated for 5 min on ice and pipetted twice after 2 min and 4 min, 10 $\times$ each, with a P200 pipette. Suspensions were passed through a 40- μ m filter into a 15-ml conical tube (each filter was rinsed with an additional 250 μ l of lysis buffer), and then transferred to a 2-ml low-binding microcentrifuge tube. Samples were centrifuged at 1,000g for 5 min at 4 °C. The supernatant was removed, leaving ~100 μ l behind, and 1 ml of wash buffer (PBS + 0.5% BSA (Miltenyi Biotec, 130-091-376) + 1 U μ l⁻¹ of RNAase inhibitor) was added with no mixing for 5 min on ice. After 5 min, the pellet was resuspended by mixing 3 $\times$ with a P1000. Samples were centrifuged again at 1,000g for 5 min at 4 °C. The supernatant was removed, leaving ~100 μ l behind, and the pellet was then resuspended in 500 μ l of wash buffer. Hoechst stain (1:10,000 dilution; Molecular Probes Hoechst 33342, Thermo Fisher, H3570) was added to each sample, and samples were then passed through 5- μ m syringe filters (pre-wetted with 500 μ l of wash buffer) using a 1-ml syringe, directly into FACS tubes. Samples were incubated for at least 5 min on ice before FACS.

FACS

Hoechst and GFP-positive nuclei were sorted using a 70- μ m nozzle and a flow rate of 3 on a BD Biosciences FACSARIA Fusion sorter into a 1.5 ml low-binding microcentrifuge tube containing collection buffer (500 μ l of 0.5% BSA + 1 U per μ l RNase inhibitor). The instrument was washed with bleach between samples. Around 150,000–200,000 nuclei per sample were sorted for further analysis.

Single-nucleus RNA-seq library preparation and sequencing

After FACS, samples were centrifuged at 1,000g for 5 min at 4 °C. The supernatant was removed and nuclei were resuspended in 20 μ l of collection buffer. The total number of nuclei for each sample was estimated (~3/4 loss) and single-nuclei suspension samples were loaded to the 10x Genomics Chromium X system using the Single Cell 3' v3.1 Reagent Kits (10x Genomics) to generate and amplify cDNA. The amplified cDNA samples were purified with Ampure XP magnetic beads (Beckman Coulter), quantified by a Qubit fluorometer (Invitrogen), and examined on a Bioanalyzer with High Sensitivity DNA chips (Agilent) for size distribution. Illumina sequencing libraries were generated from the amplified cDNA samples using the Illumina Tagment DNA Enzyme and Buffer kit (Illumina). These libraries were examined by Qubit and Bioanalyzer, then pooled at equal molar amounts and sequenced on Illumina NovaSeq 6000 S Prime flowcells as 28 + 94-nucleotide paired-end reads following the standard protocol. Raw sequencing reads were filtered by Illumina NovaSeq Control Software and only the Pass-Filter reads were used for further analysis.

Alignment and quality control of data

Alignment of reads was performed using Cell Ranger version 7.1.0. SoupX was used to remove ambient RNA contamination on the Cell Ranger output files. To subset the data that represent broken or damaged nuclei, doublets or empty droplets, we set cutoffs by checking violin plots of genes per cell: the lower bound was 150–210 features per cell for each sample and the higher bound was between 750 and 1,000 features per cell depending on the sample.

Normalization, integration and clustering

In Seurat, we used the single-cell pipeline, where single-cell transform was used for normalization. Different numbers of principal components (80, 90, 100, 110, 120, 125, 130, 150, 200) and different resolutions (0.8, 0.9, 1.0 and 1.2) were used. In total, 125 PCs at a clustering resolution of 1 was used, which resulted in 106 clusters in our dataset.

Cluster labeling/cell-type analysis

We performed cluster annotation using a combination of a hypergeometric test and AUCell algorithm as described previously⁶⁸. Detailed methods for cluster labeled are available in ref. 37.

Differentially expressed gene identification

The FindMarkers function was used in the Seurat package for differential expression identification. The Wilcoxon rank-sum test method was performed on each subsetted cluster, comparing auxin (–) cells and auxin (+) cells within the same cluster. Genes with a minimum percentage of expression in 25% of the cells were analyzed, and the significantly differentially expressed genes were identified if their log₂(fold change) > 0.1 or < –0.1, and adjusted *P* value < 0.05. The genes significantly differentially expressed in at least one cluster are considered differentially expressed. gProfiler was used to identify Gene Ontology and KEGG terms enriched among all of the hypodermal DAF-2(–) upregulated genes (FDR < 0.05) from the single-nucleus RNA-seq dataset.

Hierarchical clustering

To generate a gene expression-based hierarchical clustering dendrogram, we first aggregated the normalized gene expression level of each

gene in a cluster to obtain the average gene expression vector for each cluster, then calculated the Euclidean distance matrix between clusters, and then hierarchically clustered using the ‘hclust’ function with the ‘complete’ linkage method.

Lifespan analysis

Lifespan analyses were performed for *osm-11*-overexpressing worms. Transgenic worms or their wild-type sibling controls were segregated at the L4 larval stage to start the assay. Every other day, worms were transferred to freshly seeded NGM plates. The first day of adulthood was defined at $T = 0$. Kaplan–Meier analysis with the log-rank (Mantel–Cox) method was used to compare lifespans between transgenic and non-transgenic siblings. Worms that ‘escaped’ or ‘bagged’ were censored on the day of the event ($n = 96$ per strain; two biological replicates were performed).

Dauer assay

Day 2 adult N2 and *daf-2(e1370)* worms were allowed to lay eggs for 1 h onto OP50-seeded NGM plates at 20 °C. For intestinal DAF-2 AID worms, eggs were laid on OP50-seeded NGM plates containing 0.25% ethanol or 1 mM auxin. Adults were removed and egg-containing plates were incubated at 27 °C for 72 h. The number of dauer and non-dauer worms were counted to calculate the percentage of dauers.

Statistics and reproducibility

Two-way ANOVA with Bonferroni post hoc tests were used to compare learning indices among multiple groups. One-way ANOVAs followed by Bonferroni post hoc tests for multiple comparisons were performed. Two-way ANOVAs were used between genotype (*daf-2(e1370)* and wild-type, control RNAi and *daf-2* RNAi, control and *osm-11* overexpression, hypodermal DAF-2 AID and hypodermal DAF-2 AID;*crh-1(3315)*) and different time point (0 h, 0.5 h, 1 h, 2 h, 3 h, 4 h, 5 h and 6 h) on learning indices with a significant interaction between factors leading to the performance of Bonferroni post hoc comparisons to determine differences between individual groups. Prism 10 software was used for all statistical analyses of behavior data using standard settings for ANOVA and post hoc tests. Additional statistical details of experiments, including sample size (with n representing the number of chemotaxis assays performed for behavior, RNA collections for RNA-seq and the number of worms for microscopy), can be found in the figure legends.

Experiments were repeated on separate days, using separate independent populations, to confirm that results were reproducible, and all replicates of memory assays are provided. Software and statistical details used for RNA-seq analyses are described in the Methods. No statistical methods were used to predetermine sample sizes, but our sample sizes are similar to those reported in previous publications^{3,8,23}. Data distribution was assumed to be normal but this was not formally tested. Data collection and analysis were not performed blind to the conditions of the experiments. For feeding RNAi experiments and +/- auxin experiments, worms were randomly selected for each experiment for placement into the different experimental groups. This was achieved by randomly pipetting populations of worms into different groups from pooled starting/growth plates. No animals or data points were excluded from the analyses for any reason.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Raw bulk RNA-seq and single-nucleus RNA-seq data are publicly available on NCBI BioProject ([PRJNA1037138](https://www.ncbi.nlm.nih.gov/bioproject/PRJNA1037138)). Source data are provided with this paper. All other data are available from the corresponding author upon request.

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Author contributions

Conceptualization: S.Z., R.K. and C.T.M. Methodology: S.Z., K.E.N., Y.Z., R.K., E.T., W.Z., Y.W. and J.S.A. Investigation: S.Z., K.E.N., R.K., Y.Z. and M.E.S. Visualization: S.Z., Y.Z. and K.E.N. Funding acquisition: C.T.M. and M.-Q.D. Project administration: C.T.M. Supervision: C.T.M. Writing: S.Z., K.E.N., R.K. and C.T.M.

Competing interests

The authors declare no competing interests.

Additional information

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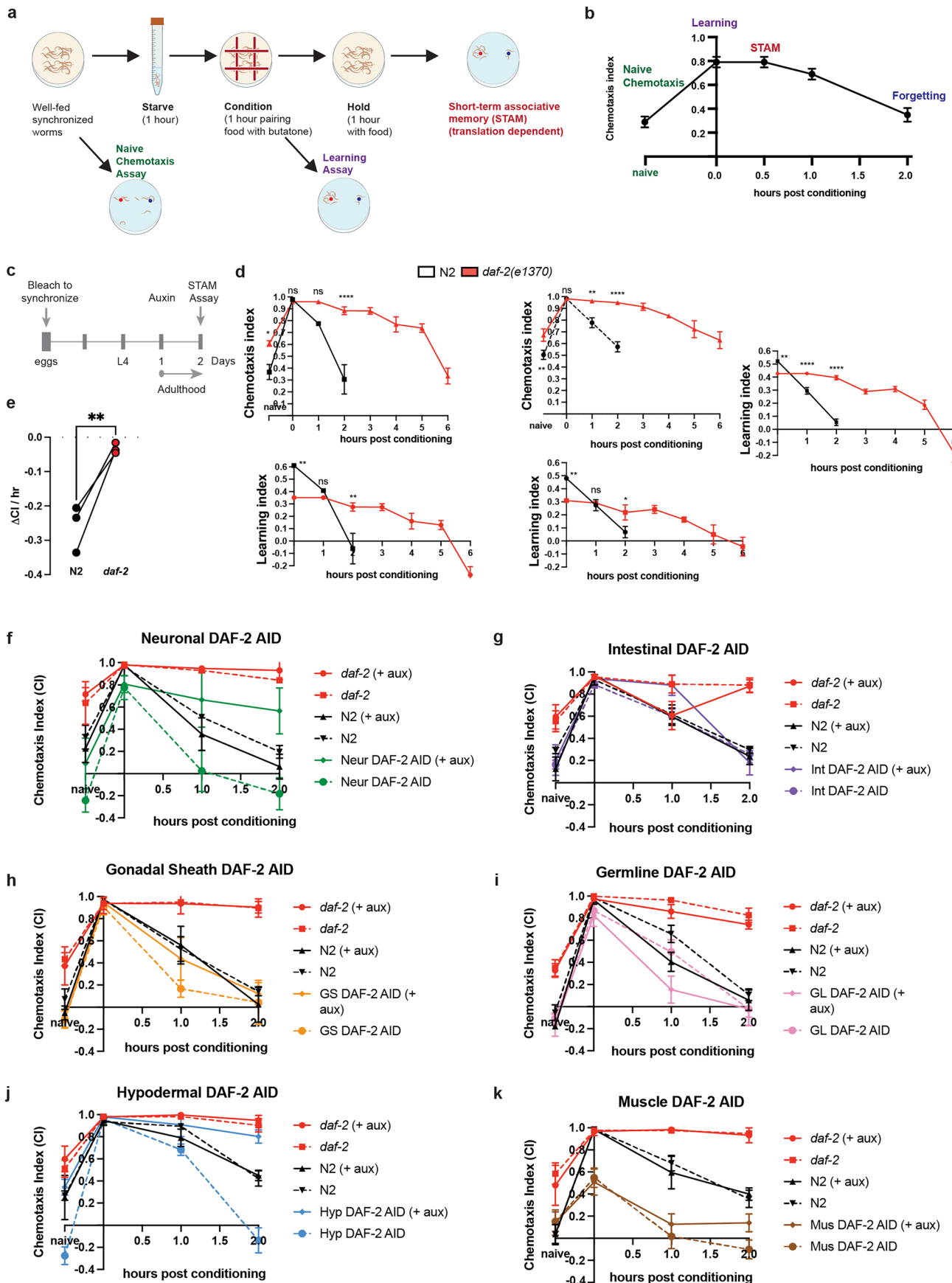
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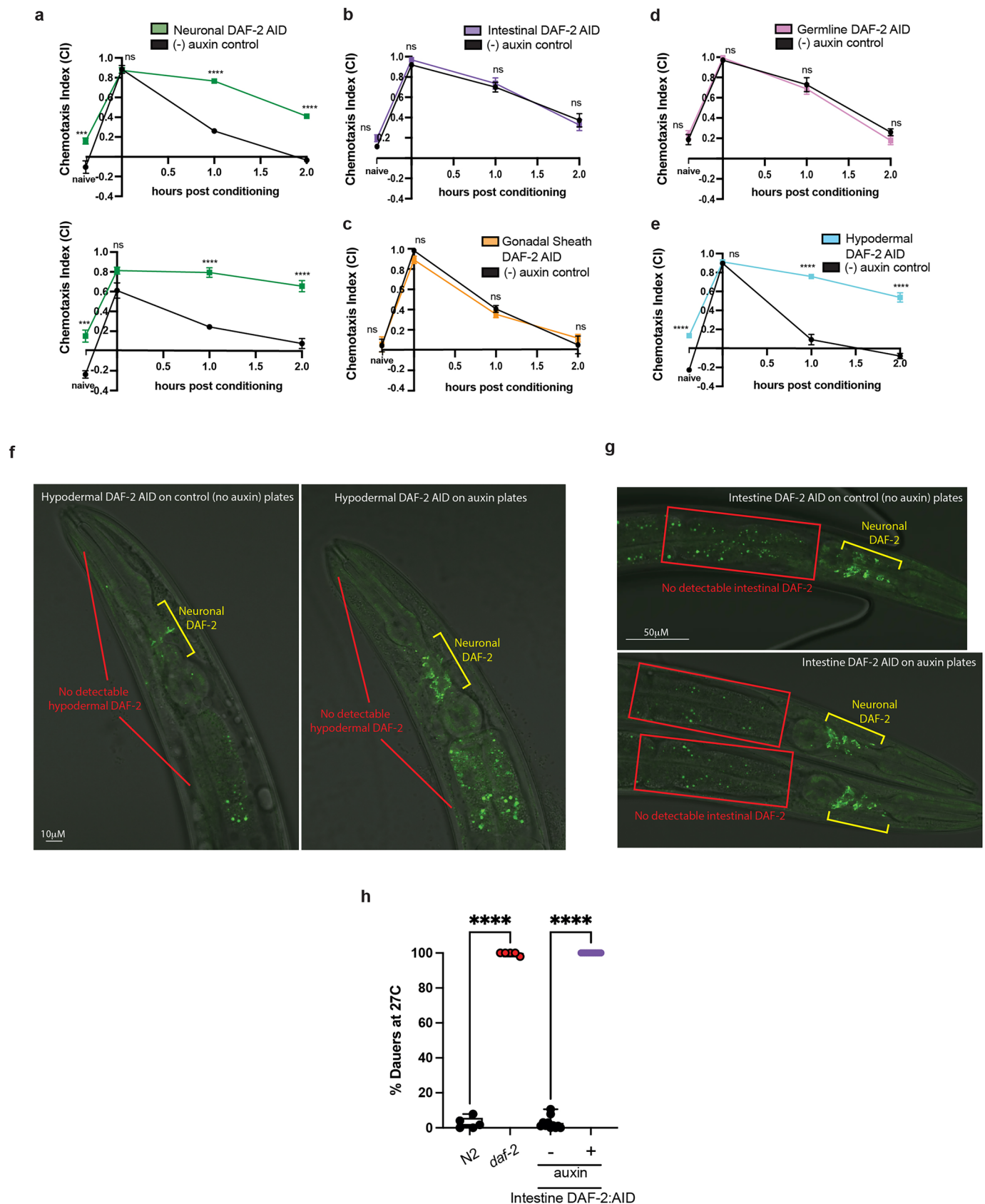
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Extended Data Fig. 1 | See next page for caption.

Extended Data Fig. 1 | Knocking down *daf-2* specifically in the hypodermis improved memory. (a) Short-term Associative Memory Assay (STAM) schematic. Learning Index is calculated by subtracting naïve chemotaxis from all subsequent timepoints. The assay distinguishes ('naïve') pre-training chemotaxis to butanone, learning ('0 timepoint,' tested immediately after training), and memory (all subsequent timepoints following training during which worms are held on food plates with no butanone). (b) Representative STAM results plotted by chemotaxis index. Timepoints representing Naïve, Learning, Short-Term Memory, and Forgetting are shown. (c) Auxin treatment is initiated on Day 1 of adulthood. Memory assays are performed on Day 2 adults following 24 h of auxin treatment and subsequent DAF-2 degradation. (d) Chemotaxis index (top) and Learning

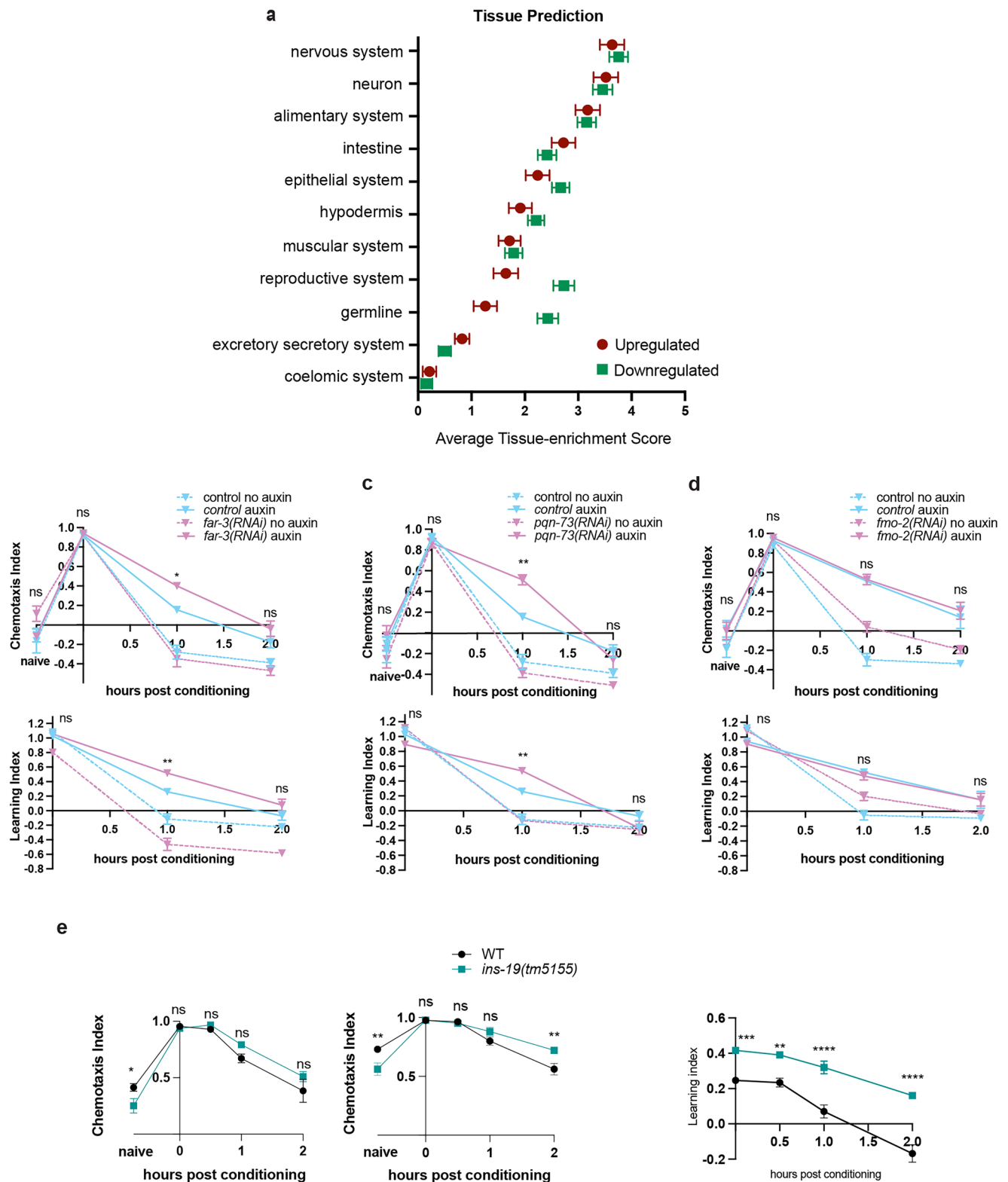
index plots (bottom) for wild-type and *daf-2* animals. All additional replicates are shown. (e) Memory was evaluated by calculating the rate of change in CI between the learning and 2 h time points across replicates. Two-tailed unpaired t-test. (f-k) Full chemotaxis indices of DAF-2 tissue-specific degradation experiments in Fig. b-f. (k) Worms with auxin-inducible DAF-2 degradation in the muscle had learning defects with or without auxin treatment. (d, f-k) Mean $\pm$ SEM. Two-way repeated measures ANOVA, Bonferroni post-hoc tests. $n \geq 4$ per time point. n represents the number of chemotaxis plates at each time point. Each plate contains ~80 worms for all memory experiments. ns = not significant, * $p \leq 0.05$, ** $p \leq 0.01$, **** $p \leq 0.0001$. All biological replicates are shown. Additional statistics provided in Supplementary Table 3. Source data available in supplement.



Extended Data Fig. 2 | See next page for caption.

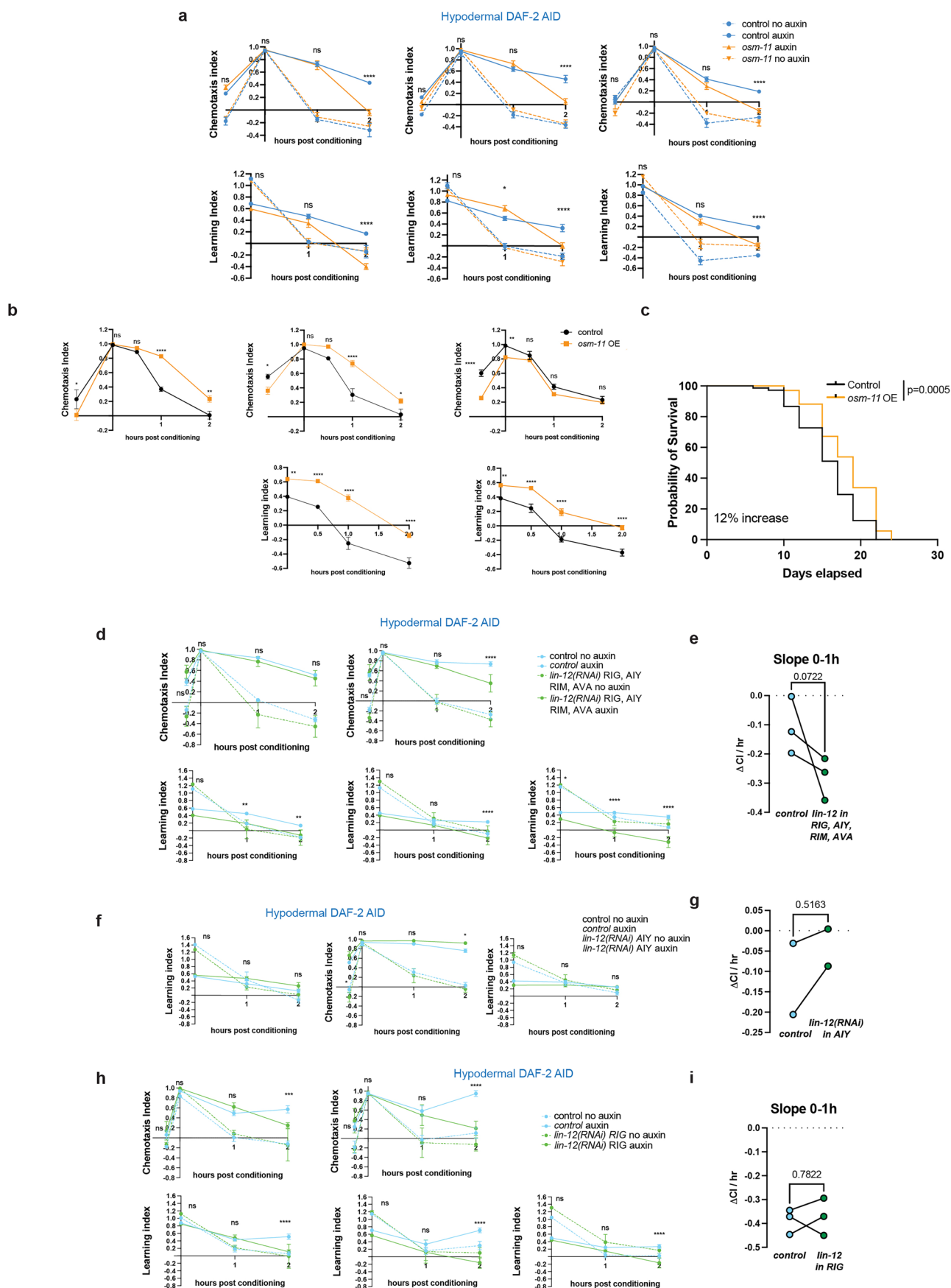
Extended Data Fig. 2 | Additional biological replicates for memory assays, and verification of intestinal DAF-2 degradation. (a-e) Biological replicates of tissue-specific DAF-2 degradation memory assays. (f) Hypodermal DAF-2 expression is too diffuse to be detected in the adult hypodermis. With and without auxin treatment conditions are shown for Day 1 hypodermal DAF-2-AID worms. Representative head images are shown. (g) DAF-2 is undetectable in the adult intestine in intestine DAF-2-AID worms with or without auxin. (h) DAF-2 degradation in the intestine was functionally verified by measuring dauer

formation with and without auxin among eggs hatched and raised at 27 °C. Two-tailed unpaired t-test. Box plots: center line, median; box range, 25–75th percentiles; whiskers denote minimum–maximum values. (a-e) Mean ± SEM. Two-way repeated measures ANOVA, Bonferroni post-hoc tests. n ≥ 4 per time point. n represents the number of chemotaxis plates at each time point. Each plate contains ~80 worms for all memory experiments. ns = not significant, *p ≤ 0.05, **p ≤ 0.01, ***p ≤ 0.001, ****p ≤ 0.0001. All biological replicates are shown. Source data available in supplement.



Extended Data Fig. 3 | Analysis of targets downstream of hypodermal DAF-2 degradation-induced memory enhancement. (a) Predicted mean tissue enrichment scores for the genes significantly altered after hypodermal DAF-2 degradation. Tissue prediction was performed using worm.princeton.edu²⁰. Mean scores $\pm$ SEM are shown for each tissue. (b–d) Chemotaxis Index plots (top) and Learning Index (bottom) plots for hypodermal DAF-2-AID worms treated with control or (b) *far-3(RNAi)*, (c) *pqn-73(RNAi)*, or (d) *fmo-2(RNAi)*. RNAi knockdown (from young adult (L4)) of the hypodermal DAF-2-regulated transcriptional

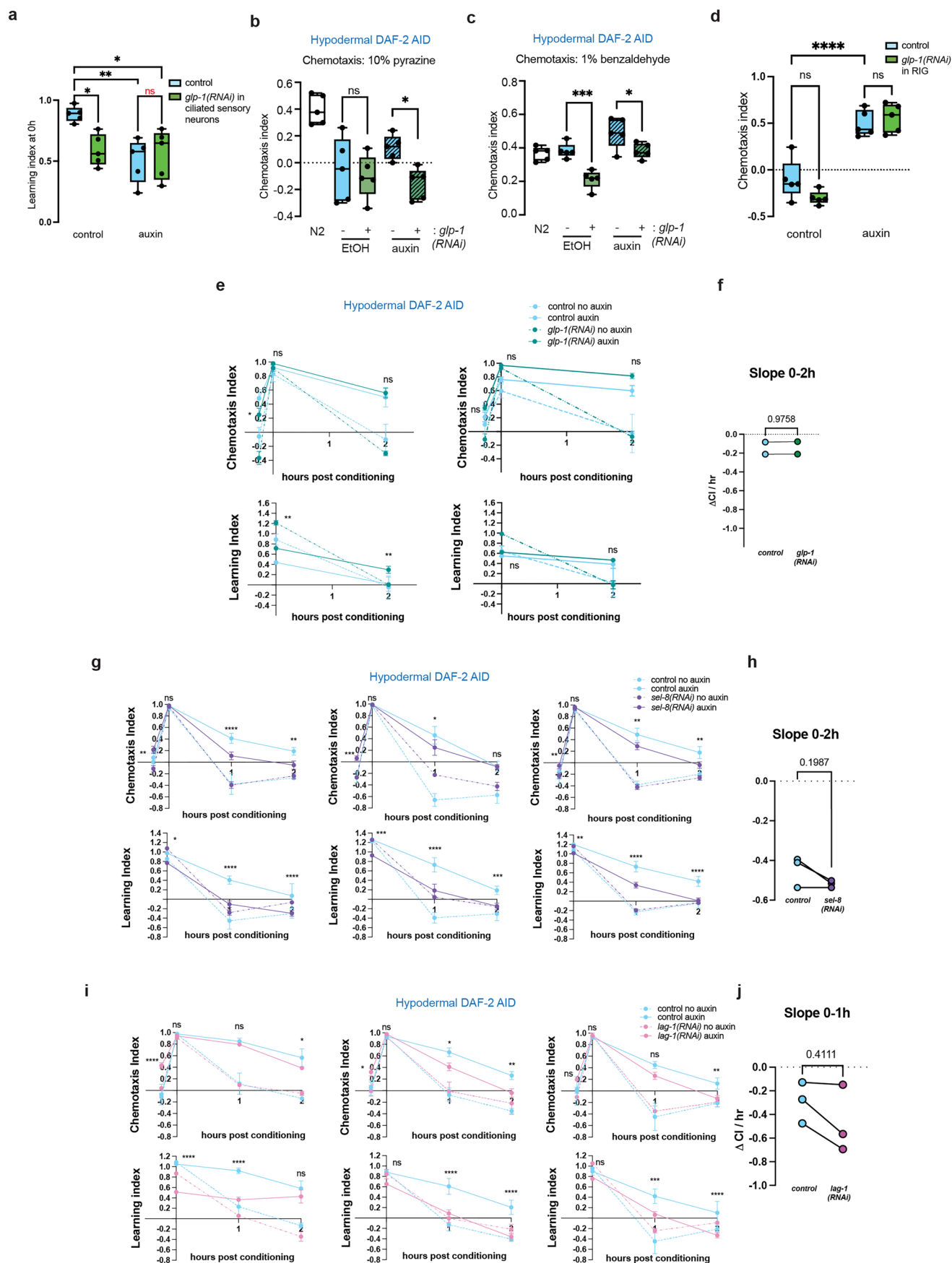
targets *pqn-73*, *fmo-2*, and *far-3* does not block memory enhancement induced by hypodermal DAF-2 degradation in young adults (tested on Day 3) (e) Chemotaxis Index plots (left) and Learning Index (right) plots for wild-type or *ins-19* mutant worms. (b–e) Mean $\pm$ SEM. Two-way repeated measures ANOVA, Bonferroni post-hoc tests. $n \geq 4$ per time point. n represents the number of chemotaxis plates at each time point. Each plate contains ~80 worms for all memory experiments. ns = not significant, * $p \leq 0.05$, ** $p \leq 0.01$, **** $p \leq 0.0001$. All biological replicates are shown. Source data available in supplement.



Extended Data Fig. 4 | See next page for caption.

Extended Data Fig. 4 | *osm-11* and Notch signaling are required for hypodermal DAF-2 degradation-induced memory enhancement. (a) Chemotaxis Index plots (top) and Learning Index (bottom) plots for hypodermal DAF-2-AID worms treated with control or *osm-11(RNAi)*. (b) Chemotaxis Index plots (top) and Learning Index (bottom) plots for wild-type or *osm-11*-overexpressing worms. (c) *osm-11* overexpression moderately increases lifespan (12%). Kaplan-Meier survival analysis. (a–b) All biological replicates shown (n = 3). (d, f, h) Chemotaxis Index plots (top) and Learning Index (bottom) plots for hypodermal DAF-2-AID worms treated with control or *lin-12(RNAi)* in (d) the *flp-18*-expressing neurons

(RIG, AIY, RIM, and AVA interneurons), (f) *ttx-3*-expressing AIY neurons, or (h) *t wk-3*-expressing RIG neurons. (e, g, i) Memory was evaluated by calculating the rate of change in CI between the learning and 1 h time points across replicates. (d–i) All biological replicates shown (n = 3). (c, e, g, i) Two-tailed unpaired t-test. (a, b, d, f, h) Two-way repeated measures ANOVA, Bonferroni post-hoc tests. Mean ± SEM. Statistical comparisons are shown only for control + auxin vs RNAi + auxin conditions. ns = not significant, *p ≤ 0.05, **p ≤ 0.01, ***p ≤ 0.001, ****p ≤ 0.0001. All biological replicates are shown. Source data available in supplement.

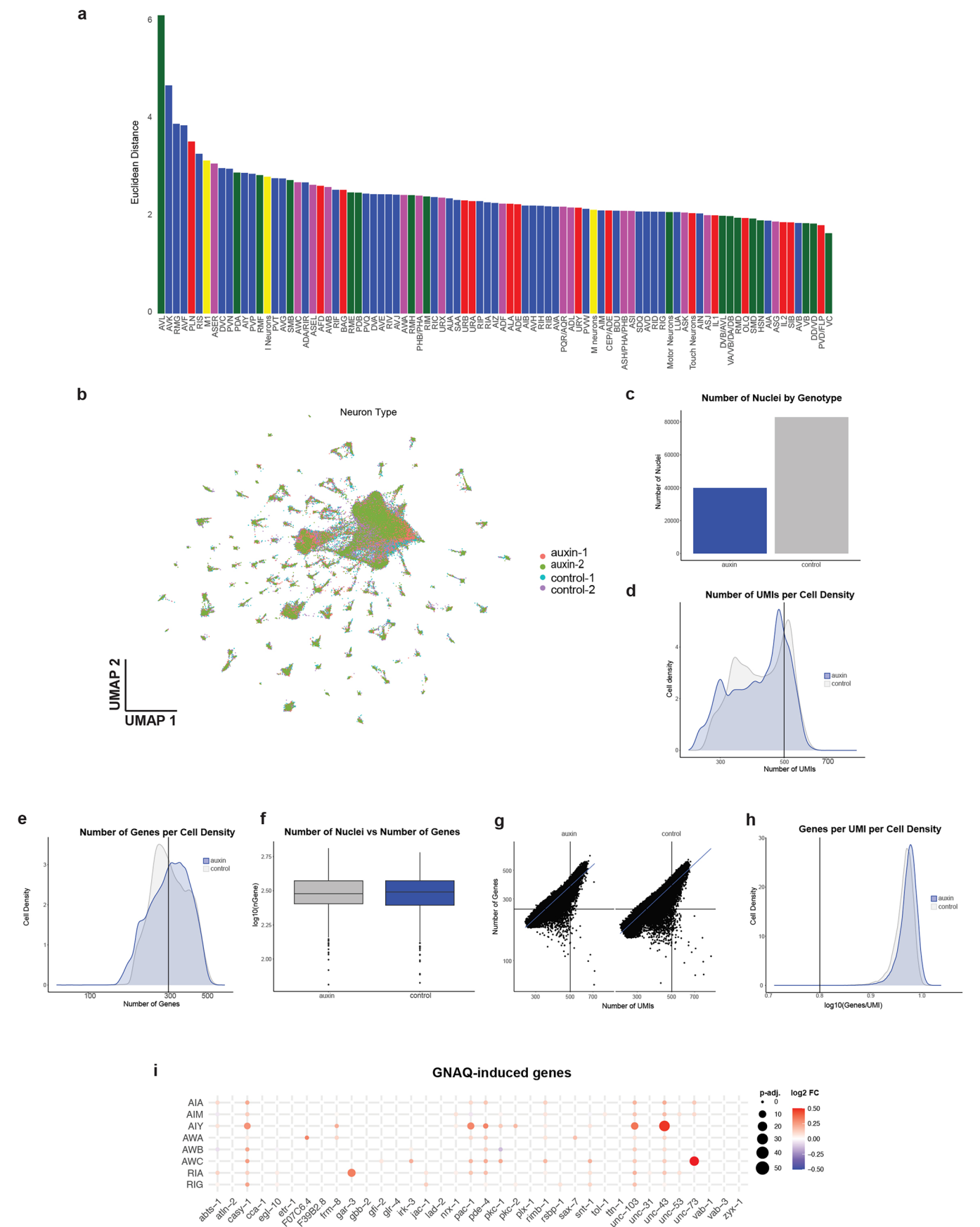


Extended Data Fig. 5 | See next page for caption.

Extended Data Fig. 5 | Behavior and memory analysis of Notch signaling genes.

(a) Knocking down *glp-1* specifically in ciliated neurons impairs learning ability. 0 h time point is shown. (b, c) Knocking down *glp-1* in the ciliated neurons impairs chemotaxis towards (b) 10% pyrazine and (c) 1% benzaldehyde in young Day 2 adult worms. (d) Knocking down *glp-1* specifically in the RIG does not disrupt memory enhancement. No defect in motility or chemotaxis was observed. (e) Chemotaxis Index plots (top) and Learning Index (bottom) plots for hypodermal DAF-2-AID worms *glp-1(RNAi)* in the RIG neurons. (e, g, i) Chemotaxis Index plots (top) and Learning Index (bottom) plots for hypodermal DAF-2-AID

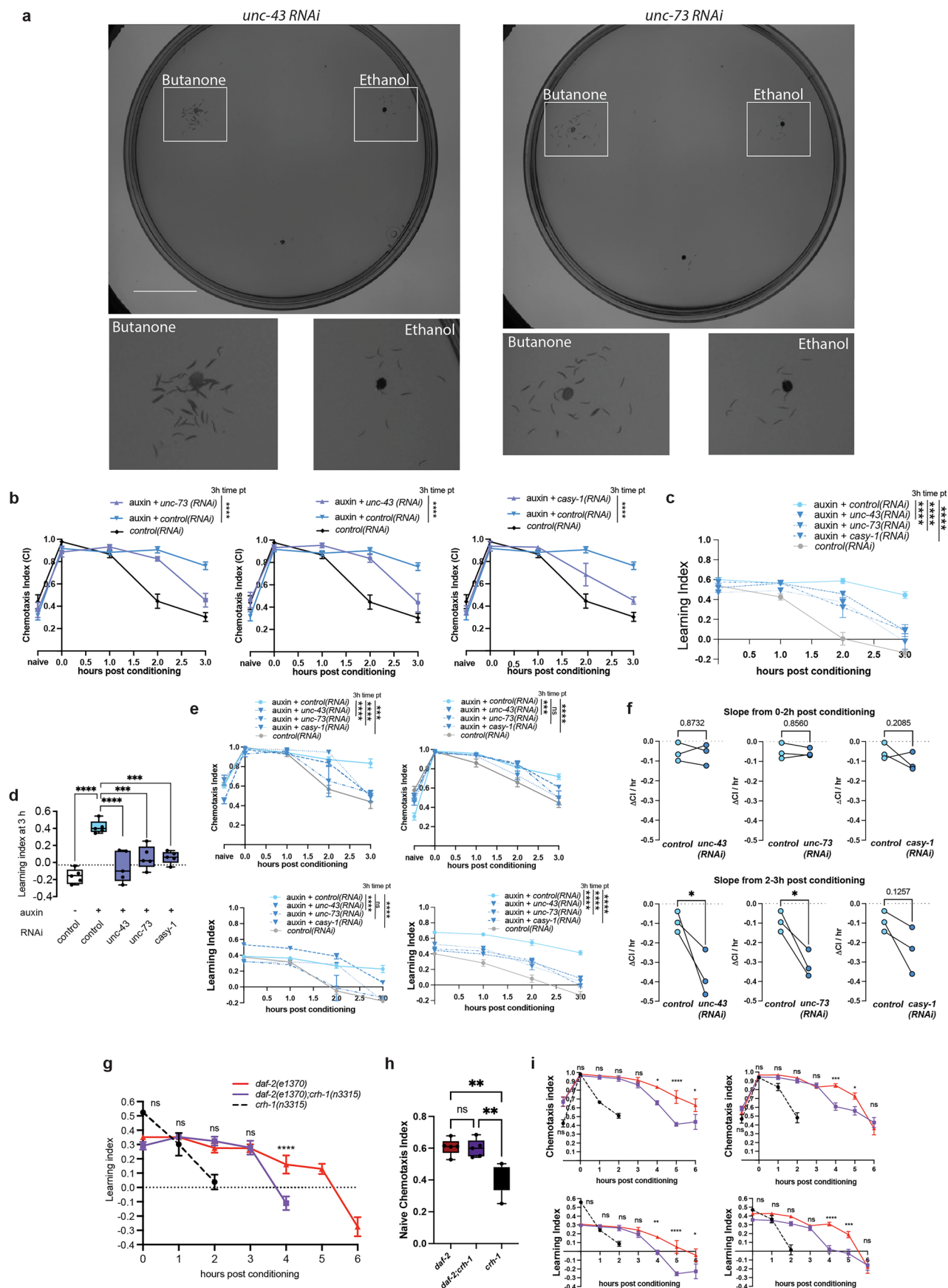
worms treated with control or (e) *glp-1(RNAi in RIG)*, (g) *sel-8(RNAi)*, or (i) *lag-1(RNAi)*. (f, h, j) Memory was evaluated by calculating the rate of change in CI between the learning and indicated time points across replicates. (a-d) Two-way repeated measures ANOVA, Tukey's post-hoc tests. Mean $\pm$ SEM (e, g, i) Two-way repeated measures ANOVA, Bonferroni post-hoc tests. Mean $\pm$ SEM. Statistical comparisons are shown only for control + auxin vs RNAi + auxin conditions. (a-d) $n = 3$ biological replicates. (e-j) All biological replicates shown. (f, h, j) Two-tailed unpaired t-test. ns = not significant, * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$, **** $p \leq 0.0001$. All biological replicates are shown. Source data available in supplement.



Extended Data Fig. 6 | See next page for caption.

Extended Data Fig. 6 | Analysis of single-nucleus sequencing from hypodermal DAF-2-AID worms treated with or without auxin. (a) Euclidean distance of each neuron's mean Control (no auxin) vector and mean hypodermal DAF-2(-) vector. (b) UMAP of all nuclei color coded by sample type and biological replicate. (c) Number of nuclei by genotyping. (d) Number of UMIs per cell density. (e) Number of genes per cell density. (f) Number of nuclei relative to the number of genes auxin: minima 1.83, maxima 2.81, centre 2.48, bounds of box and whiskers 2.40,

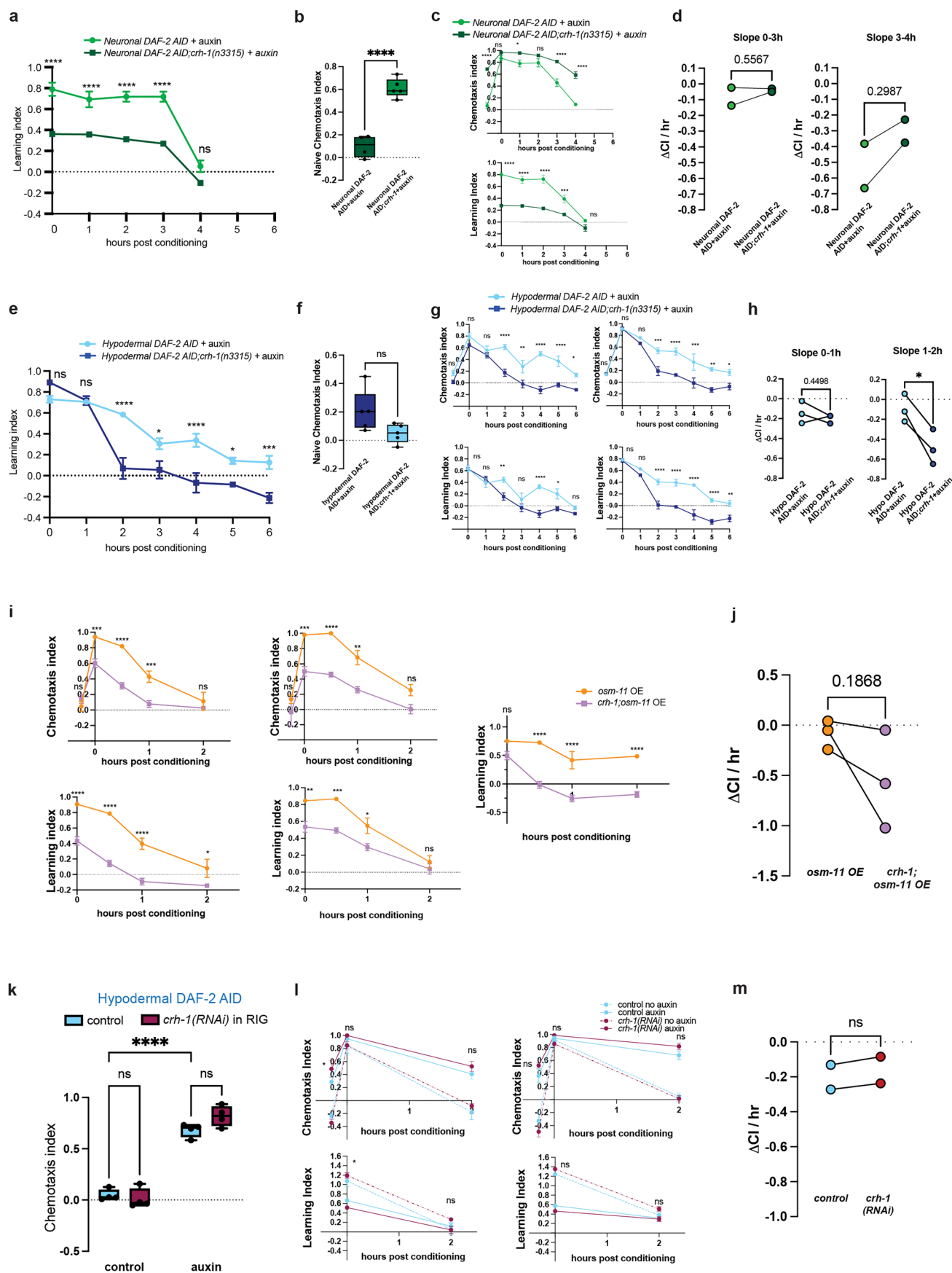
2.57 and percentile (10 – 2.35) (90 – 2.64). Control: minima 1.83, maxima 2.78, centre 2.49, bounds of box and whiskers 2.39, 2.57 and percentile (10 – 2.31) (90 – 2.63). Representative graph for n = 2 biological replicates for single-nucleus sequencing. (g) Number of genes relative to the number of UMIs. (h) Genes per UMI per cell density. (i) Hypodermal DAF-2(-) upregulated GNAQ-induced-genes in select neurons.



Extended Data Fig. 7 | See next page for caption.

Extended Data Fig. 7 | CREB-downstream genes are required for hypodermal DAF-2 degradation-induced memory. (a) RNAi of *unc-43* and *unc-73* RNAi worms does not disrupt motility or chemotaxis. Inset images show worms that have successfully migrated from the origin (bottom dot) to the odorant or control spot. (b, c) Chemotaxis Index plots (b) and Learning Index (c) for each condition shown in Fig. 5e. (d) Learning Index at 2 h for control, *unc-43*, *unc-73*, or *casy-1* RNAi in the neuron-RNAi-sensitized hypodermal DAF-2 AID strain. (e) Additional biological replicates shown for chemotaxis Index plots (top) and Learning Index (bottom) plots for hypodermal DAF-2 AID worms treated with control or *unc-43*, *unc-73*, or *casy-1* RNAi. (f) Memory was evaluated by calculating the rate of change

in CI between the learning indicated time points across replicates. (g) Learning index plot for data shown in Fig. 5f. (h) Naïve index plot for *daf-2*, *daf-2;crh-1*, and *crh-1* worms. One-way ANOVA, Bonferroni correction. (i) Additional biological replicates shown for *daf-2*, *daf-2;crh-1*, and *crh-1* memory assays. All biological replicates shown (n = 3). (g, i) Statistical comparisons are shown for *daf-2* vs *daf-2;crh-1*. (b-e, g, i) Two-way ANOVA, Bonferroni post-hoc tests. Mean ± SEM. (f) Two-tailed unpaired t-test. Box plots: center line = median, box range is 25th - 75th percentile; whiskers denote minimum-maximum values. *p ≤ 0.05, **p ≤ 0.01, ***p ≤ 0.001, ****p ≤ 0.0001. All biological replicates are shown. Source data available in supplement.

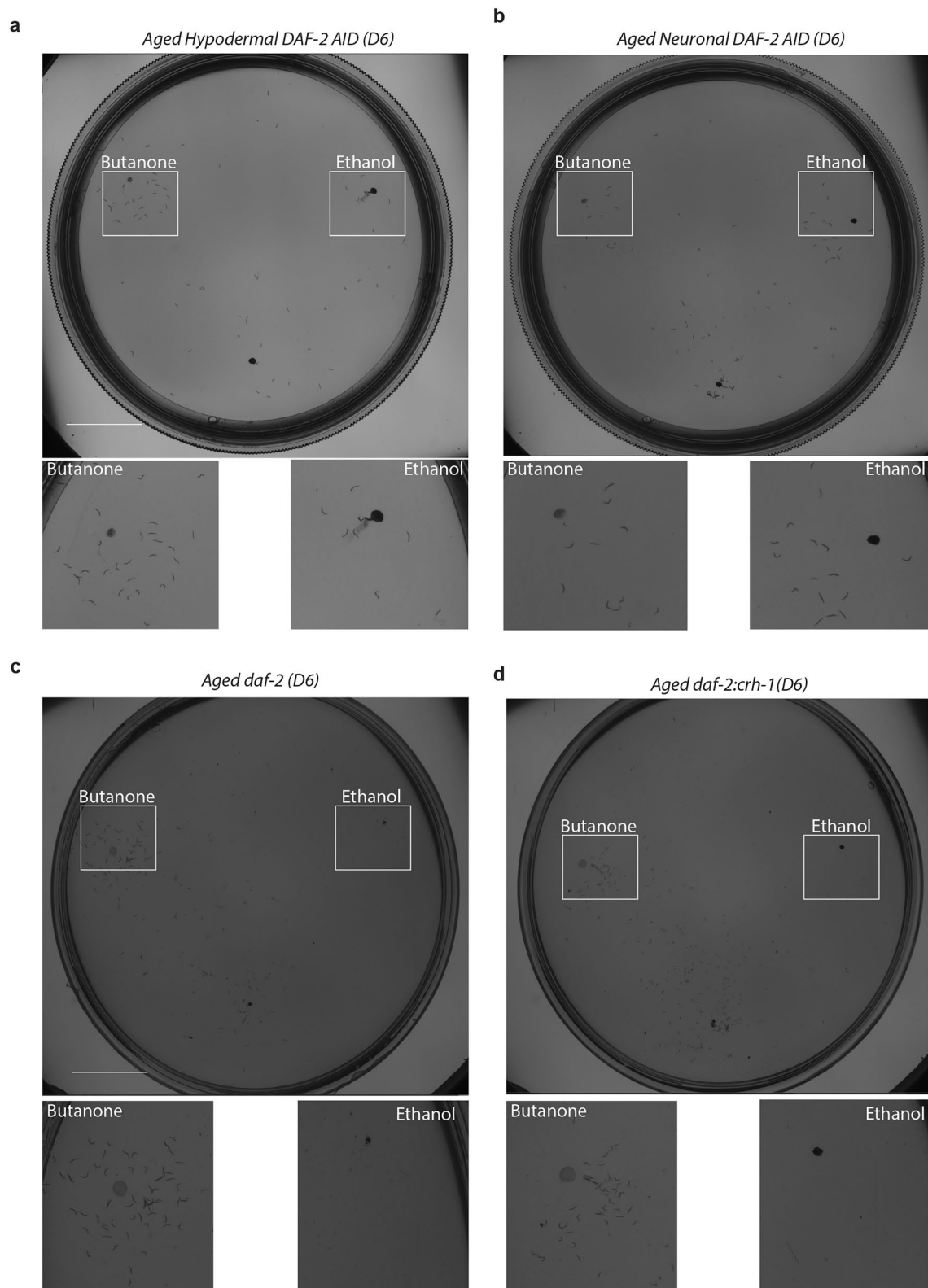


Extended Data Fig. 8 | See next page for caption.

Extended Data Fig. 8 | CREB is required hypodermal DAF-2(-) memory.

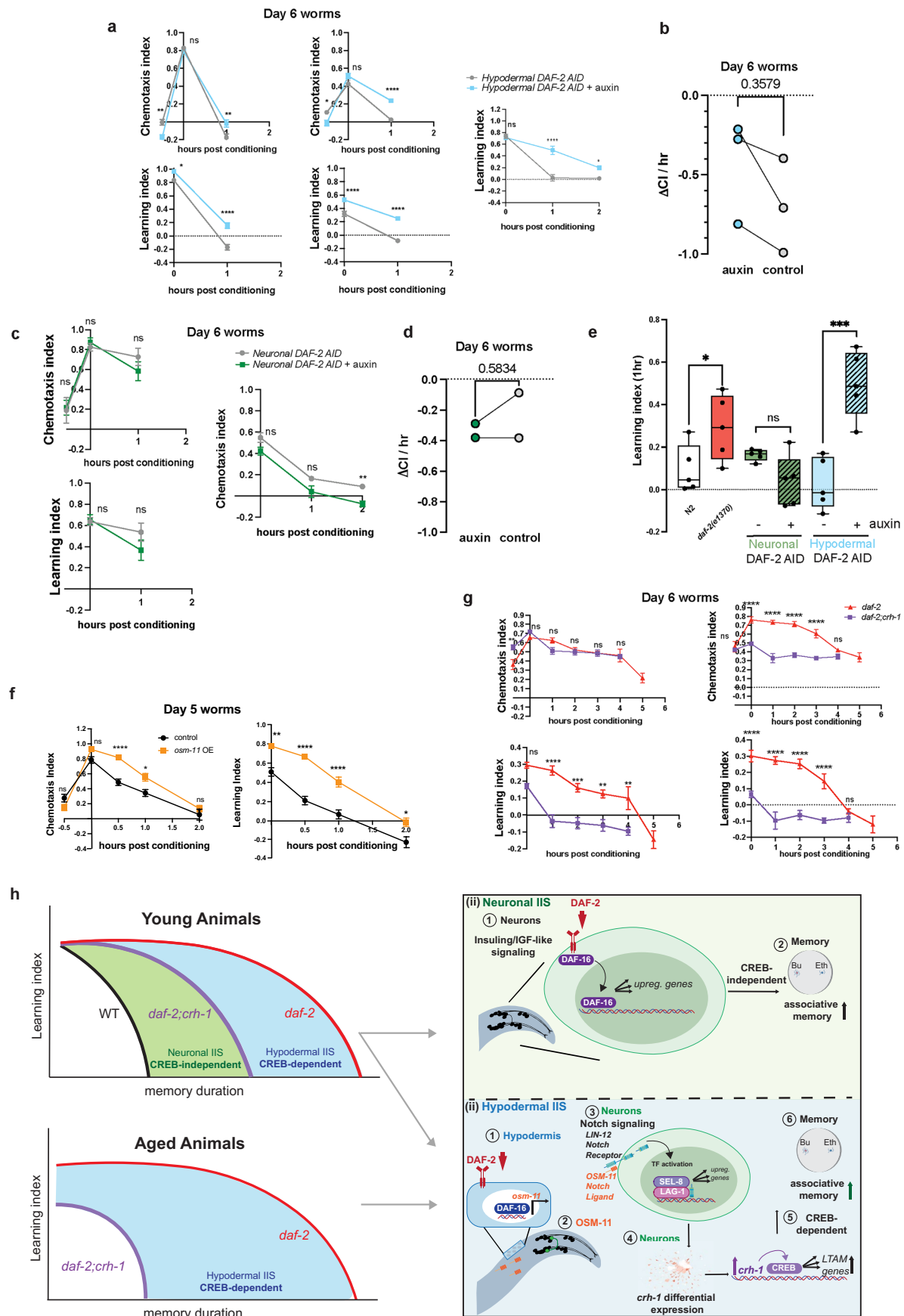
(a) Learning index and (b) naïve chemotaxis index for neuronal DAF-2 AID worms $\pm$ auxin. (c) Additional biological replicates are shown for neuronal DAF-2 AID worms $\pm$ auxin. (d) Memory was evaluated by calculating the rate of change in CI between the learning and indicated time points across replicates. (e) Learning index and (f) naïve chemotaxis index for hypodermal DAF-2 AID worms $\pm$ auxin. (g) Additional biological replicates are shown for hypodermal DAF-2 AID worms $\pm$ auxin. (h) Memory was evaluated by calculating the rate of change in CI between the learning and indicated time points across replicates. (i) Chemotaxis Index plots (top) and Learning Index (bottom) plots for *osm-11 OE* or *osm-11 OE;crh-1* worms. (j) Memory was evaluated by calculating the rate of

change in CI between the learning and 0.5 h time points across replicates. (k) Disruption of *crh-1* specifically in RIG (*twk-3p::crh-1(RNAi)*) does not impair memory in hypodermal DAF-2(-) worms. Two-way ANOVA, Tukey's test. (l) CI and LI plots, and additional biological replicates are shown for the experiment shown in (k). (m) Memory was evaluated by calculating the rate of change in CI between the learning and 2 h time points across replicates. (a, c, e, g, i, l) Two-way ANOVA, Bonferroni post-hoc tests. Mean $\pm$ SEM. $n \geq 4$. (b, d, f, h, j, m) Two-tailed unpaired t-test. Box plots: center line = median, box range is 25th - 75th percentile; whiskers denote minimum-maximum values. ns = not significant, * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$, **** $p \leq 0.0001$. All biological replicates are shown. Source data available in supplement.



Extended Data Fig. 9 | Motility is intact in aged worms. Motility and chemotaxis of aged (Day 6) worms is not impaired in hypodermal or neuronal DAF-2 AID worms or *daf-2* and *daf-2;crh-1* mutants. **a–d**, Aged (Day 6) hypodermal DAF-2 AID worms (**a**), aged (Day 6) neuronal DAF-2 AID worms (**b**), aged (Day 6) *daf-2* worms

(**c**) and aged (Day 6) *daf-2;crh-1* worms (**d**) show normal motility and chemotaxis. Inset images show worms that have successfully migrated from the origin (bottom dot) to the odorant or control spot.



Extended Data Fig. 10 | See next page for caption.

Extended Data Fig. 10 | Hypodermal DAF-2 degradation improves memory with age similar to *daf-2* mutants. (a) Chemotaxis Index plots (top) and Learning Index (bottom) plots for hypodermal Day 6 DAF-2-AID worms treated with or without auxin. (b) Memory was evaluated by calculating the rate of change in CI between the learning and 1 h time points across replicates. (c) Chemotaxis Index plots (top) and Learning Index (bottom) plots for Day 6 neuronal DAF-2-AID worms treated with or without auxin. (d) Memory was evaluated by calculating the rate of change in CI between the learning and 1 h time points across replicates. (e) 1 h memory in Day 6 worms was compared across wild-type worms, *daf-2* mutants, and neuronal and hypodermal DAF-2 AID worms $\pm$ auxin. Two-tailed unpaired t-test. Box plots: center line = median, box range is 25th - 75th percentile; whiskers denote minimum - maximum values. (f) Additional biological replicates shown for Day 6 control or *osm-11*-overexpressing worms. (g) Additional biological replicates shown for Day 6 *daf-2* and *daf-2;crh-1* worms. (a, c, e, f, g) Mean $\pm$ SEM. Two-way repeated measures ANOVA, Bonferroni post-hoc tests. $n \geq 4$ per time point. n represents the number of chemotaxis plates at each time point. Each plate contains ~80 worms for all

memory experiments. ns = not significant, * $p \leq 0.05$, ** $p \leq 0.01$, **** $p \leq 0.0001$. (b, d, g, Two-tailed unpaired t-test. All biological replicates are shown. Source data available in supplement. h, A Model of Non-autonomous regulation of memory through Hypodermal IIS and Notch signaling to neuronal CREB activity. Young animals exhibit memory extension after a single training session that is partially CREB-dependent, and partially CREB-independent. The neuronal-IIS pathway controls CREB-independent, DAF-16-dependent gene expression that regulates CREB-independent short-term associative memory. Aged animals exhibit memory extension after a single training session that is CREB-dependent. The reduction of DAF-2 in the hypodermis induces *osm-11* gene expression; OSM-11 signals to neurons and activates neuronal Notch signaling by binding to the Notch receptor LIN-12. The activated Notch receptor is cleaved^{33,69,70}, and the intracellular domain transfers into the nucleus^{71,72}, forming a complex with the transcription factor LAG-1 and the coactivator SEL-8 to regulate downstream Notch target gene transcription⁷³ (for example, *ins-19*). CREB transcriptional activation of memory genes contributes to hypodermal DAF-2 degradation-induced associative memory.

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- ☐ ☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
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Only common tests should be described solely by name; describe more complex techniques in the Methods section.
- ☐ ☒ A description of all covariates tested
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- ☐ ☒ For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
Give P values as exact values whenever suitable.
- ☒ ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
- ☒ ☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
- ☐ ☒ Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated

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Software and code

Policy information about [availability of computer code](#)

Data collection

Nikon eclipse Ti
Bioanalyzer 2100
Illumina NovaSeq 6000 S Prime

Data analysis

g:Profiler (version e109_eg56_p17_1d3191d)
Nikon NIS elements software (v5.42.03)
Illumina NovaSeq Control Software
Cutadapt (v1.6)
RNA STAR (v2.7.8a)
htseq (v0.9.1)
samtools (v1.7)
DESeq2 (v1.22.1)
Prism 10
Seurat (v4.1.3)
Cell Ranger (v7.1.0)
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- Accession codes, unique identifiers, or web links for publicly available datasets
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Raw bulk RNA-seq and single-nucleus RNA-seq data are publicly available on NCBI BioProject (PRJNA1037138). Source data for memory assays are available in Table Sx. Microscopy files are available from the lead contact upon request.

Human research participants

Policy information about [studies involving human research participants and Sex and Gender in Research](#).

Reporting on sex and gender

n/a

Population characteristics

n/a

Recruitment

n/a

Ethics oversight

n/a

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size

No statistical methods were used to predetermine sample size. Sample sizes were determined for each experiment based on the standard for previously published and optimized assays (For lifespan assays, Murphy et al. Nature (2003). For short-term associative memory assays, Kauffman et al, 2010; Lakhina et al, 2015; Kaletsky et al, 2016; and Arey et al, 2018. For dauer assay, Zhang et al, 2022. For bulk RNA-sequencing assays, Kaletsky et al, 2016; Kaletsky et al, 2018. For single-nucleus RNA sequencing, St. Ange et al, 2024.

Data exclusions

No data were excluded from the study.

Replication

All experiments were replicated using independent biological experiments (performed/collected on different days). At least 2-3 biological replicates were performed. All attempts at replication were successful.

Randomization

For feeding-RNAi experiments and +/- auxin experiments that comprise all of the experiments in the paper, worms were randomly selected for each experiment for placement into the different experimental groups. This was achieved by randomly pipetting populations of worms into different groups from pooled starting/growth plates.

Blinding

Blinding was not performed. This was due to the need to maintain proper treatment conditions for each strain throughout the experiments.

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☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Animals and other research organisms

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Laboratory animals	Wild type, mutant, and transgenic <i>C. elegans</i> were used throughout the study. Specific mutant strains are specified in the Methods section under ' <i>C. elegans</i> Strains'. Ages of animals are noted in the figure legends.
Wild animals	No wild animals were used in the study.
Reporting on sex	<i>C. elegans</i> hermaphrodites were exclusively used in this study for memory assays. Males were used in the study for mating purposes to generate crosses, as noted in the Methods section.
Field-collected samples	No field-collected samples were used in the study.
Ethics oversight	No ethical oversight was required since <i>C. elegans</i> are invertebrates.

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